



Bharati Vidyapeeth

(Deemed to be University)



Institute of Management and Entrepreneurship
Development, Pune

A REPORT FOR MINOR PROJECT - I ON

“KloudPg-Online Pg Booking ”

IN THE COMPLETE FULFILLMENT OF THE REQUIREMENT FOR DEGREE
OF

MASTER OF COMPUTER APPLICATIONS

(SEMESTER II)

SUBMITTED BY

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Certificate of Completion

This is to certify that **Aaditya Ray Rajput (PRN No: 2503100208)**, **Aishwarya Bajirao Ghadage (PRN No: 2503100221)**, **Aditya Sanjay Gaikwad (PRN No: 2503100220)**, **Rohan Kale (PRN No: 2503100229)**, is a bona-fide students of Semester-II, Master of Computer Application (MCA) degree program of Bharati Vidyapeeth (Deemed to be University), Pune for the academic year **2025–2027**.

As a part of the University curriculum, we have successfully completed our minor project - I titled “**Online Pg Booking**” under the guidance of **Ms. Deepti Deshmukh**.

This project work fulfills the requirement of the university course module “MINOR PROJECT - I” of Semester-II.

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DECLARATION

We, the undersigned, hereby declare that the project report entitled “Deployment of **“KloudPg Online Pg Booking”** is developed and submitted by us in partial fulfillment of the requirement for the award of the degree of Master of Computer Application (MCA), Semester-II, for the academic year 2025–2027 under the guidance of **Ms. Deepti Deshmukh**.

This project report is based on our original work and has not been submitted previously either in part or in full, for the award of any other degree or diploma. All the information included in this report has been collected, analyzed and presented by us during the course of the project work.

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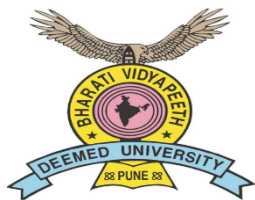
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Date:

Place:



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ACKNOWLEDGEMENT

Preparing this acknowledgement is not a merely a formality, but an opportunity for us to express our sincere gratitude to all those who have supported us throughout the completion of this mini project.

We express our deep sense of gratitude to my project guide, **Ms. Deepti Deshmukh**, for his valuable guidance, continuous support, and encouragement throughout the project work. His insightful suggestions and constant motivation helped us to successfully complete this project.

We would like to express our sincere thanks to our PYTHON mentor, Dr. Satyawan Hembade, Faculty member at **BVDU (IMED), Pune**, for his clear teaching, insightful suggestions during the project topic selection, and continuous encouragement greatly helped us in successfully completing our learning and project work.

I would like to thank all the faculty members of **Bharati Vidyapeeth (Deemed to be University), Pune**, for their support and encouragement.

Lastly, we express our heartfelt gratitude to our family and friends for their constant support and motivation throughout the completion of this project.

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1. INTRODUCTION

1.1 About the Project

The **Online PG Booking and Management System** is a web-based application designed to bridge the gap between accommodation seekers and property owners. The primary goal is to simplify the complex process of searching, booking, and managing Paying Guest (PG) facilities through a centralized digital platform.

Key Objectives include:

- **Streamlined Search:** To provide a fast and efficient system for users to find PGs based on specific locations and GPS data.
- **Digital Transactions:** To enable secure online room booking and integrated payment processing to replace manual cash transactions.
- **Operational Efficiency:** To empower PG owners with tools to manage listings, update room availability in real-time, and handle pricing dynamically.
- **Centralized Oversight:** To provide an administrative panel for monitoring system activities, managing user verification, and generating performance reports.
- **Enhanced Communication:** To improve the user experience by providing built-in chat features and automated notifications for booking confirmations.

1.2 Client/Requirement Description

The system is designed to cater to three distinct stakeholder groups, each with specific requirements to ensure a transparent rental ecosystem:

- **Customers (Students and Working Professionals):** This group requires a user-friendly interface to register, filter PGs by facilities and rent, view property images/videos, and track their booking history.
- **PG Owners:** This group requires a robust management dashboard to list properties, manage room inventory, respond to booking requests, and communicate directly with potential tenants.
- **Administrators:** The system requires a high-level control module to approve or reject PG listings, moderate user reviews, block malicious accounts, and resolve complaints to maintain system integrity.

- **System Integrity:** The project must ensure data privacy, secure session management, and a responsive interface that functions across different network conditions.

2. COMPANY/PROJECT PROFILE

The Online PG Booking and Management System is developed as an academic project under the Master of Computer Applications (MCA) program. This project simulates a real world software solution aimed at solving common problems faced in the PG (Paying Guest) accommodation sector.

The system represents a startup-like digital platform that provides online services for searching, booking, and managing PG accommodations. It is designed to act as an intermediary between customers (tenants) and PG owners, ensuring smooth communication and efficient transaction handling.

This project focuses on delivering a scalable and user-friendly web application that incorporates modern technologies such as:

- Frontend: React ,Vite, React Router Dom
- Backend: Node.js
- Database: MongoDB

The platform provides:

- A customer interface for searching and booking PGs
- An owner dashboard for managing listings and bookings
- An admin panel for monitoring and controlling system activities

The system is designed with the goal of:

- Improving accessibility to PG accommodations
- Reducing manual effort in booking and management
- Ensuring transparency and reliability in the rental process

Although developed as an academic project, this system has the potential to be expanded into a real-world commercial application with additional features such as mobile app integration, advanced search filters, and AI-based recommendations.

3. SYSTEM ANALYSIS

3.1 Existing System

The current process for finding and managing Paying Guest accommodations is largely manual and fragmented.

- **Manual Search:** Customers often have to physically visit multiple locations or rely on word-of-mouth and local advertisements to find available PGs.
- **Lack of Transparency:** There is no centralized way to verify room details, pricing, or facilities before visiting, leading to wasted time and effort.
- **Offline Transactions:** Payments and bookings are handled through cash or verbal agreements, which lack a digital trail and security.
- **Communication Gap:** Direct communication between owners and potential tenants is difficult until a physical meeting occurs.
- **Data Management:** Owners and admins manage records using paper-based systems or basic spreadsheets, making it difficult to track real-time availability or generate reports.

3.2 Proposed System

The proposed **Online PG Booking and Management System** is a web-based solution designed to automate the entire rental lifecycle.

- **Centralized Platform:** Provides a single interface where customers can search for PGs based on specific locations using GPS.
- **Real-time Updates:** Owners can update room availability, pricing, and property images instantly, ensuring customers see accurate data.

- **Integrated Booking & Payment:** Users can reserve rooms and make secure online payments directly through the platform.
- **Role-Based Access:** Specialized dashboards for Customers, Owners, and Admins to handle their specific tasks efficiently.
- **Automated Notifications:** The system sends real-time alerts for booking confirmations and updates to keep all stakeholders informed.

3.3 Feasibility Study

The feasibility of the system is assessed across three primary dimensions to ensure successful implementation:

- **Technical Feasibility:** The system is developed using standard web technologies like HTML, CSS, and JavaScript for the frontend, with a robust backend using Node.js and MongoDB/MySQL. These technologies are well-supported and scalable.
- **Economic Feasibility:** By digitizing the process, the system reduces the need for physical advertisements and manual record-keeping. The cost of hosting and maintenance is low compared to the efficiency gains for both owners and customers.
- **Operational Feasibility:** The interface is designed to be user-friendly, requiring minimal technical knowledge from customers or PG owners. It addresses the clear demand for a more transparent and convenient way to handle PG rentals.
- **Schedule Feasibility:** The project can be completed within the given academic timeline. • Development can be done in phases • Testing and deployment are manageable within schedule Hence, the project is feasible within the given time constraints.

4. PROJECT REQUIREMENT SPECIFICATION

4.1 Module Description

The system is divided into several functional modules to ensure a modular architecture and ease of maintenance. These modules handle specific tasks ranging from identity verification to complex booking logic and administrative oversight.

- **Authentication Module:** Manages secure registration, login, and session handling for all user roles (Customer, Owner, Admin).
- **Search & Filtering Module:** Utilizes location data and GPS to allow users to find nearby accommodations based on specific criteria.
- **Property Management Module:** Enables PG owners to upload property details, images, videos, and manage room availability.
- **Booking & Payment Module:** Facilitates room selection, date management, and secure financial transactions.
- **Communication Module:** Features an integrated chat system and notification engine to keep users and owners connected.
- **Admin Dashboard:** Provides high-level control over user verification, listing approvals, and complaint resolution.

4.2 Coding Standards

To maintain high-quality software development and ensure the system is scalable, the following standards are followed:

- **Naming Conventions:** Clear and descriptive names for variables, functions, and classes (e.g., `calculateTotalRent`, `PGController`).
- **Modularization:** Code is organized into reusable components and controllers to prevent redundancy.

- **Commentary:** Extensive use of inline comments to explain complex logic, particularly in database schemas and payment integration.
- **Security Standards:** Implementing secure password hashing, session management, and protecting against common vulnerabilities like SQL injection or NoSQL injection.
- **Version Control:** Regular commits using Git to track changes and facilitate collaboration within the project group.

4.3 Module 1 Description: User Authentication & Profile

This module serves as the entry point for all users of the system.

- **Registration:** Collects user details such as name, email, and role (Customer or Owner) to create an account.
- **Secure Login:** Validates credentials against the database and maintains the user session.
- **Password Management:** Allows users to reset or update their passwords securely.
- **Profile Management:** Enables users to view and update their personal information and track their activity history.

4.4 Module 2 Description: PG Search & Booking

This is the core functional module for the customer-facing side of the platform.

- **Location-Based Search:** Uses GPS or text input to filter properties by specific geographical areas.
- **Property View:** Displays comprehensive details including rent, room type, available facilities (Wi-Fi, food, etc.), and media galleries.
- **Reservation Logic:** Allows customers to select specific dates and rooms, checking real-time availability before proceeding.
- **Transaction Processing:** Integrates with a payment gateway to handle secure deposits or full rent payments.

5. SYSTEM REQUIREMENTS

5.1 Software Requirements

The software stack for the **Online PG Booking and Management System** is selected to ensure cross-platform compatibility and efficient data handling.

- **Operating System:** Windows 10/11, Linux (Ubuntu), or macOS for development and hosting.
- **Frontend Technologies:** HTML5, CSS3, and JavaScript are used to build a responsive user interface.
- **Backend Environment:** Node.js for server-side logic and API management.
- **Database:** MongoDB (NoSQL) or MySQL (Relational) for storing user profiles, property listings, and booking data.
- **Development Tools:** VS Code (Editor), Git (Version Control), and Postman (API Testing).
- **Web Browser:** Modern browsers like Google Chrome, Mozilla Firefox, or Microsoft Edge for accessing the application.

5.2 Hardware Requirements

The hardware specifications are designed to support smooth performance for both the server and the end-user.

- **Processor:** Intel Core i3 or higher (or equivalent AMD processor) for development and local hosting.
- **RAM:** Minimum 4GB (8GB recommended) to handle concurrent database queries and server processes.
- **Hard Disk:** At least 500MB of free space for application files, media uploads (images/videos), and database storage.
- **Internet Connectivity:** Stable broadband connection required for GPS integration, payment gateway processing, and real-time notifications.
- **Display:** Standard monitor with a minimum resolution of 1024x768 for the administrative and owner dashboards.

6. SYSTEM DESIGN

6.1 Detailed Design

The architecture follows a modular approach. Below is the mapping of core user stories to system classes:

User Story	Main Class
Register/Login	User / AuthController
Search PG	PG / SearchController
Booking & Payment	Booking / Payment
Admin Control	Admin / AdminController

6.1.1 Overview of UML Diagrams

Unified Modeling Language (UML) diagrams are the standard visual language used to document the architecture and design of the **Online PG Booking and Management System**. These diagrams provide a clear blueprint for developers and stakeholders by representing the system from multiple perspectives.

The diagrams used in this project help in:

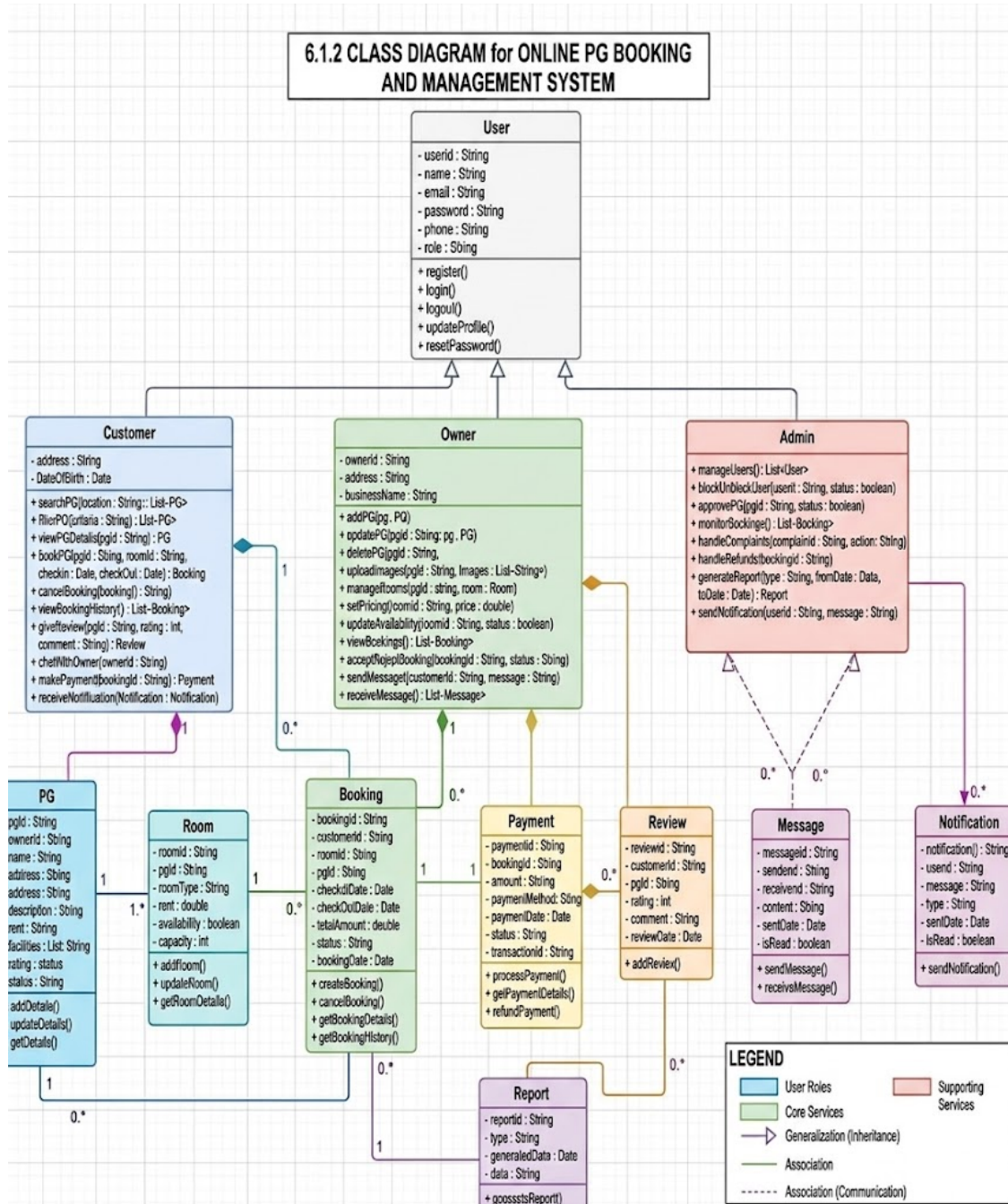
- **Understanding System Design:** Providing a high-level view of how different modules connect.
- **Improving Communication:** Ensuring developers and clients have a shared understanding of system functionality.
- **Simplifying Complexity:** Breaking down complex processes, like payment and booking, into manageable visual steps.

The project utilizes four primary types of UML diagrams:

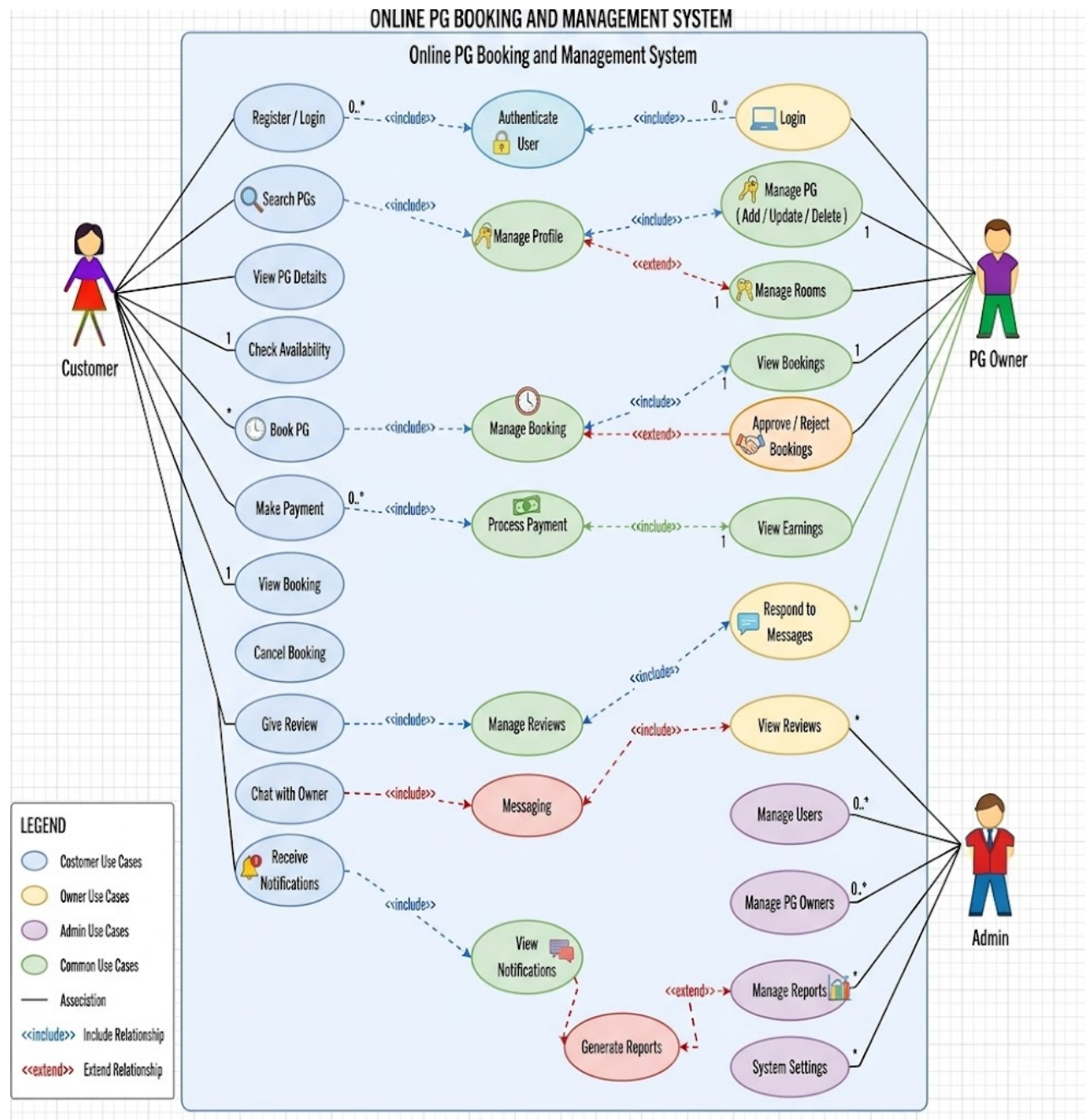
1. **Class Diagram:** A structural diagram that defines the system's objects, their properties (attributes), and what they can do (methods). For example, it shows how a customer is a type of User.

2. **Use Case Diagram:** A functional diagram that identifies the interactions between users (Actors) and the system. It shows "who" does "what," such as a **customer** searching for a PG or an **Owner** adding a listing.
3. **Activity Diagram:** A behavioral diagram that acts like a flowchart. It maps the step-by-step logic of specific workflows, such as the sequence of events from clicking "Book" to receiving a "Confirmation".
4. **Component Diagram:** An architectural diagram that represents the physical parts of the system. It shows how the **Frontend interface**, the **Node.js Backend**, and external **APIs** (like payment gateways) are organized and connected.

6.1.2 Class Diagram

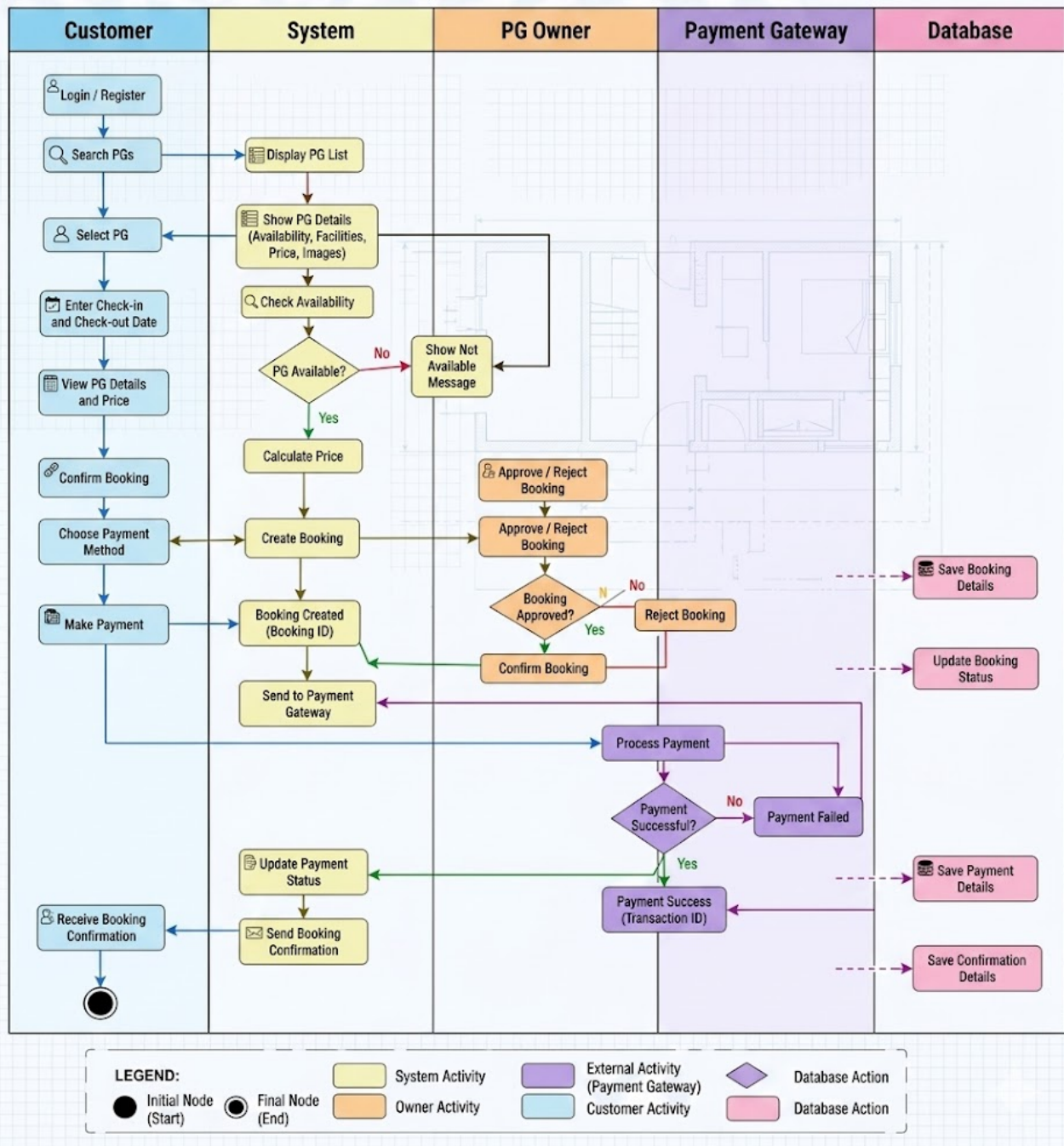


6.1.3 Use Case Diagram

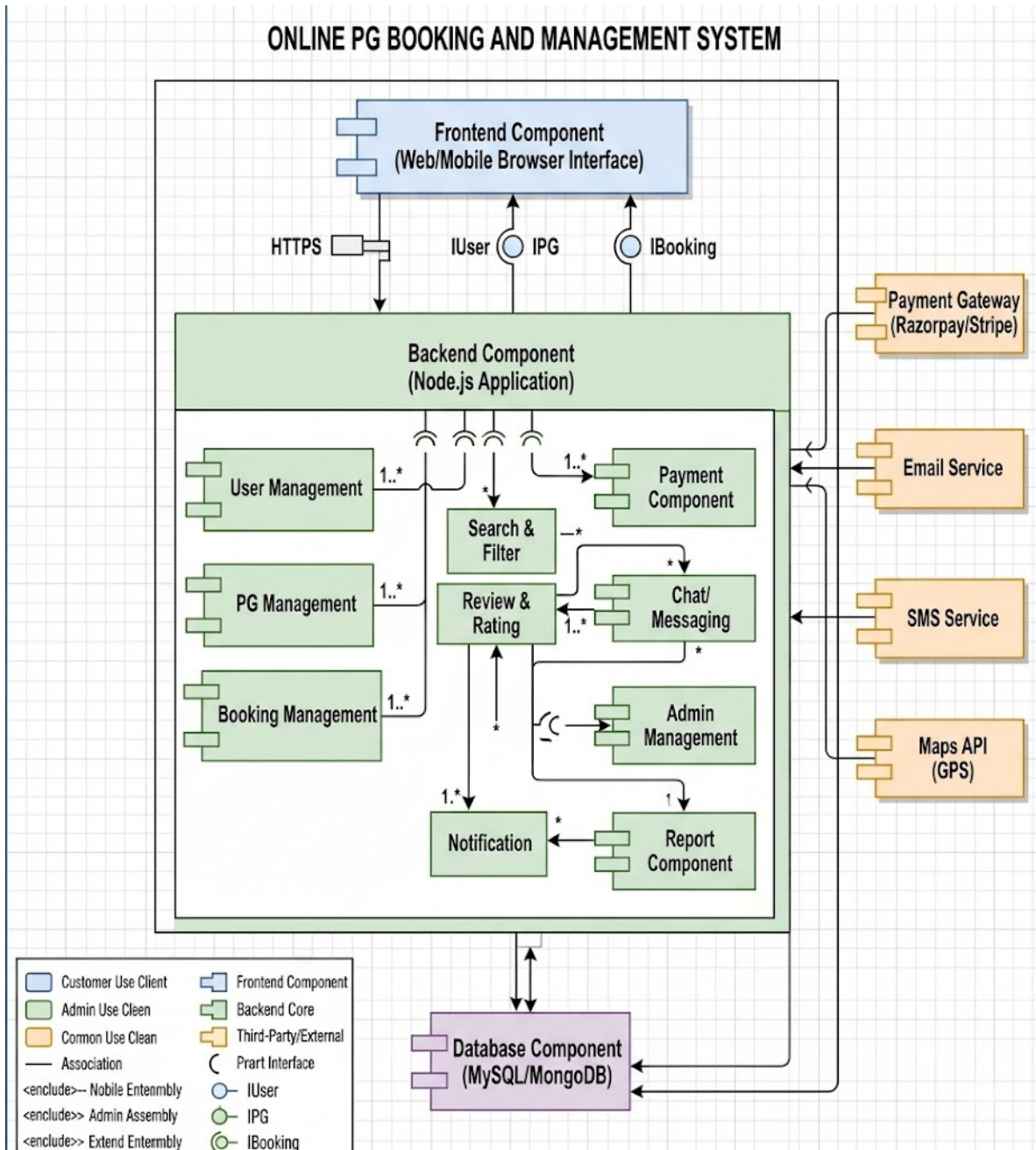


6.1.4 Activity Diagram

6.1.4 ACTIVITY DIAGRAM - ONLINE PG BOOKING AND MAKE PAYMENT



6.1.5 Component Diagram



6.2 DATABASE DESIGN

The database design for the **Online PG Booking and Management System** ensures that all property, user, and transaction data is stored efficiently and securely. This phase focuses on data integrity, normalization, and clear documentation of the system's storage architecture.

6.2.1 DB STANDARDS

To maintain a professional and scalable database, the following standards are implemented:

- **Naming Conventions:** All table names are plural (e.g., Users, Bookings) and written in snake_case.
- **Primary Keys:** Every table includes a unique identifier, typically id or [entity]_id, as a primary key.
- **Data Normalization:** The database follows Third Normal Form (3NF) to eliminate data redundancy and ensure logical dependencies.
- **Consistency:** Consistent data types are used across tables (e.g., UUID for identifiers and TIMESTAMP for dates).

6.2.2 DATA DICTIONARY

The data dictionary serves as a centralized repository of metadata, defining the meaning and characteristics of each data element:

Entity	Attribute	Description
User	email	Unique login identifier for the user.
PG	rent	Monthly cost of the PG accommodation.
Booking	status	Current state of the reservation (Pending, Confirmed, Cancelled).
Payment	tx_id	Unique transaction reference from the payment gateway.

6.2.3 TABLE STRUCTURE

The system is built upon a relational structure. Key tables include:

- **Users Table:** Stores credentials and profile details for Customers, Owners, and Admins.
- **PGs Table:** Contains property descriptions, location coordinates, and owner associations.
- **Rooms Table:** Tracks individual room details within a PG, including type (Single/Sharing) and availability.
- **Bookings Table:** Links users to rooms with check-in and check-out dates.

6.2.4 LOW LEVEL DESIGNING

Low-level design (LLD) involves the actual implementation details of the database schema. This includes defining specific SQL data types, constraints (Not Null, Unique), and indexes for optimized searching.

In this phase, we map the **Class Diagram** relationships into physical database foreign keys. For example, the ownerId in the PGs table acts as a foreign key referencing the userId in the Users table, ensuring that every property is linked to a valid owner.

7. TEST CASES

To ensure the **Online PG Booking and Management System** is thoroughly validated, the testing process is divided into specific modules. This approach ensures that each functional unit operates correctly before being integrated into the larger system.

Test ID	Module	Input	Expected Output
TC-01	Auth	Valid credentials	Successful Login
TC-02	Booking	Date selection	Room reserved correctly
TC-03	Admin	Report Request	PDF report generated

1. User Authentication & Profile Module

This module focuses on secure access and personal data management for all three user roles.

- **TC_AUTH_01:** Verify registration with a duplicate email address. **Expected:** System prevents registration and displays a "User already exists" error.
- **TC_AUTH_02:** Test the "Forgot Password" functionality with a valid email. **Expected:** System sends a reset link or OTP to the registered email.
- **TC_AUTH_03:** Verify that a user cannot access the dashboard without logging in (Session Management). **Expected:** User is redirected to the Login page.

2. PG Management Module (Owner & Admin)

This module handles the property lifecycle, including listing creation, image uploads, and verification.

- **TC_PG_01:** Owner attempts to add a PG with missing mandatory fields (e.g., rent or address). **Expected:** Form validation triggers and highlights missing fields.

- **TC_PG_02:** Verify image upload functionality for property photos. **Expected:** Images are stored correctly and displayed in the property preview.
- **TC_PG_03:** Admin reviews a new property listing. **Expected:** Status changes from "Pending" to "Approved" or "Rejected" based on Admin action.

3. Search & Booking Module (Customer)

This is the core functional module where customers interact with listings and initiate stay requests.

- **TC_BK_01:** Test search filters (e.g., Price Range, WiFi, Food availability). **Expected:** Results strictly match the selected criteria.
- **TC_BK_02:** Verify that a user cannot book a room for past dates. **Expected:** Date picker restricts past dates, or system returns a "Invalid Date" error.
- **TC_BK_03:** Attempt to book a room that has already reached maximum capacity. **Expected:** System displays "Room Full" and disables the booking button.

4. Payment & Transaction Module

This module handles the critical financial integration with external gateways.

- **TC_PAY_01:** Verify the calculation of total amount including security deposit and taxes. **Expected:** The final price matches the predefined logic in the system.
- **TC_PAY_02:** Test payment with an expired credit card on the payment gateway simulator. **Expected:** System displays "Transaction Failed" and maintains the booking as "Pending".
- **TC_PAY_03:** Verify that a successful payment generates a digital receipt and unique Transaction ID. **Expected:** Receipt is accessible in the "My Bookings" section.

5. Communication & Notification Module

This module ensures real-time updates between the Customer and the Owner.

- **TC_NOT_01:** Verify that the Owner receives a notification immediately after a customer book a PG. **Expected:** Push or in-app notification appears in the Owner's dashboard.

- **TC_NOT_02:** Test the chat functionality between Customer and Owner. **Expected:** Messages are delivered and stored in chronological order.

6. Admin Reports & Analytics Module

This module focuses on the business intelligence and oversight features.

- **TC_REP_01:** Generate a monthly revenue report for a specific property. **Expected:** System compiles all successful payments for that month into a downloadable PDF/CSV.
- **TC_REP_02:** Verify the "User Management" panel's ability to deactivate a fraudulent account. **Expected:** The deactivated user can no longer log in or access system features.

1. User Authentication & Profile

Test Case ID	Test Scenario	Expected Result
TC_01	Register with valid details and unique email.	Account created successfully; user redirected to login.
TC_02	Login with incorrect password.	System displays "Invalid Credentials" error message.
TC_03	Update user profile (phone number/address).	Changes are saved in the database and reflected on the profile page.

2. PG Discovery & Booking

Test Case ID	Test Scenario	Expected Result
TC_04	Search for PGs by location (e.g., "Pune").	System displays a list of PGs available in that specific area.
TC_05	Check availability for a room on specific dates.	System accurately shows "Available" or "Booked" based on DB status.
TC_06	Submit a booking request for an available room.	Booking status set to "Pending"; notification sent to PG Owner.

3. Payment Processing

Test Case ID	Test Scenario	Expected Result
TC_07	Complete payment using a valid test card via Razorpay.	Transaction successful; booking status updates to "Confirmed".
TC_08	Close payment gateway window before transaction ends.	System handles the interruption and marks payment as "Failed/Pending".

4. Owner & Admin Management

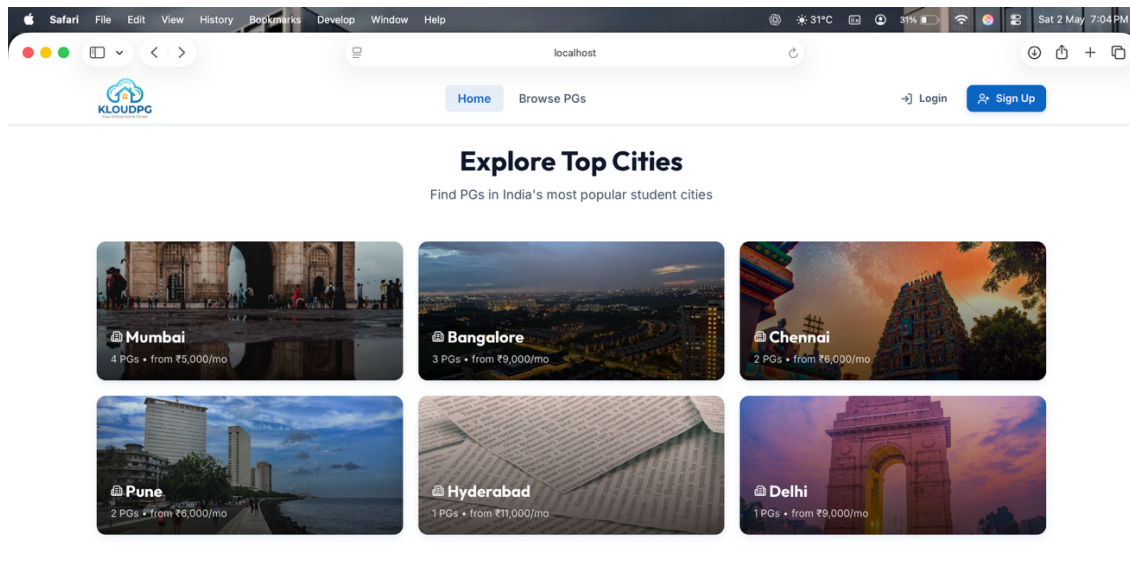
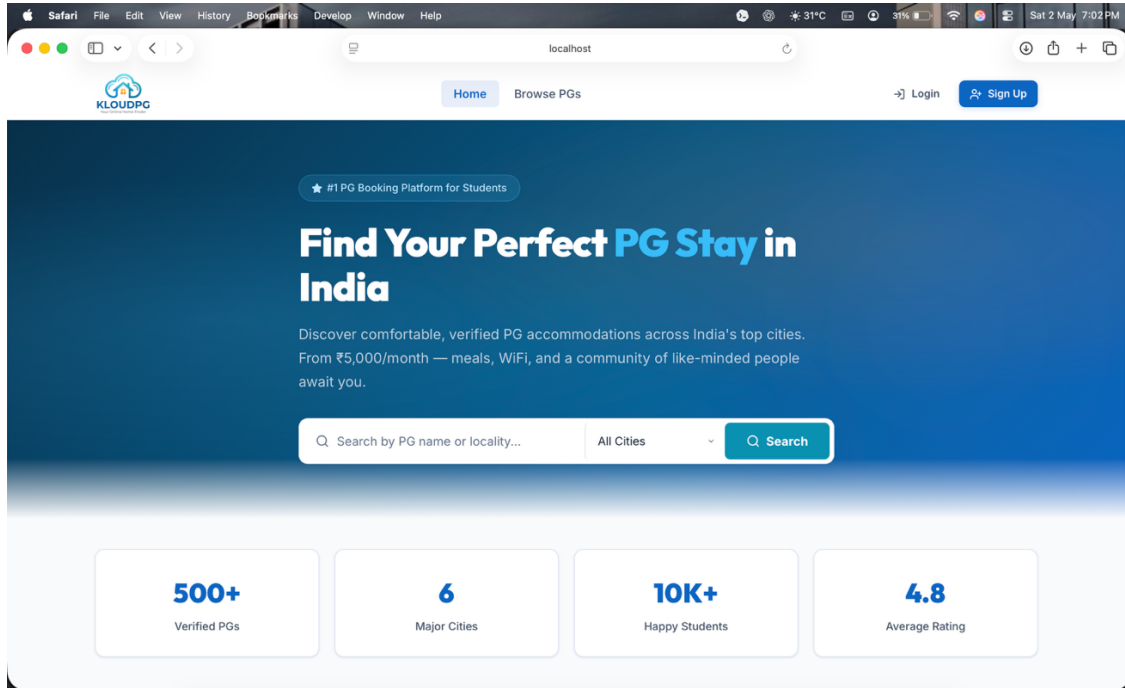
Test Case ID	Test Scenario	Expected Result
TC_09	PG Owner adds a new property with images and rent.	Property appears in the admin's "Pending Approval" queue.
TC_10	Admin approves a new PG listing.	The PG becomes visible and searchable for all Customers.
TC_11	Owner rejects a booking request.	Booking status updates to "Rejected"; refund process (if applicable) triggers.

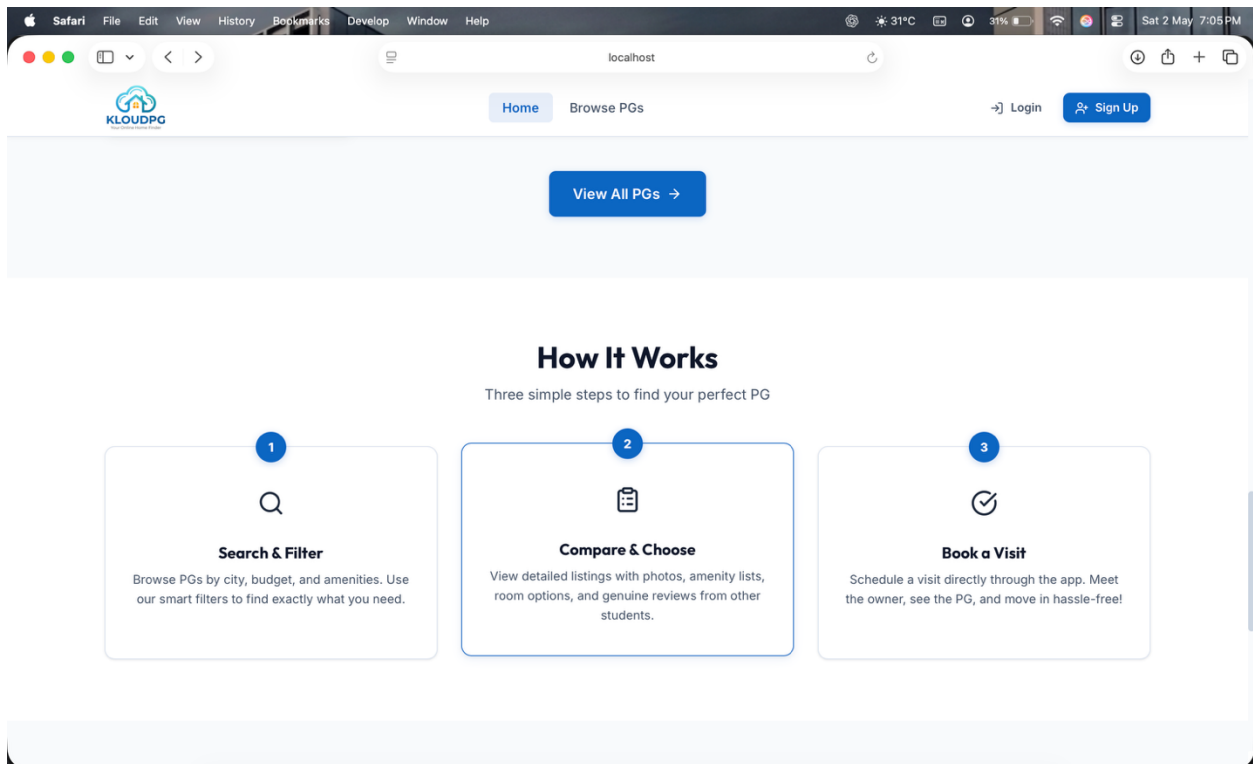
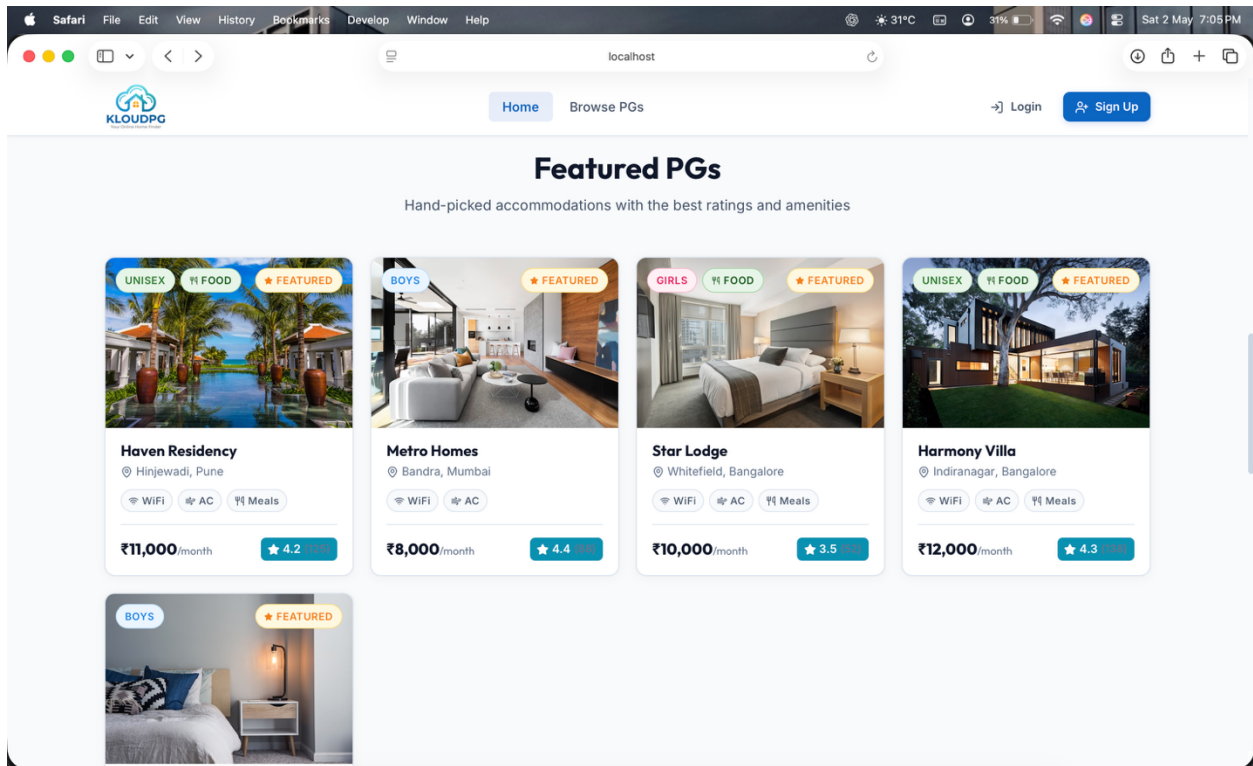
TEST SUMMARY

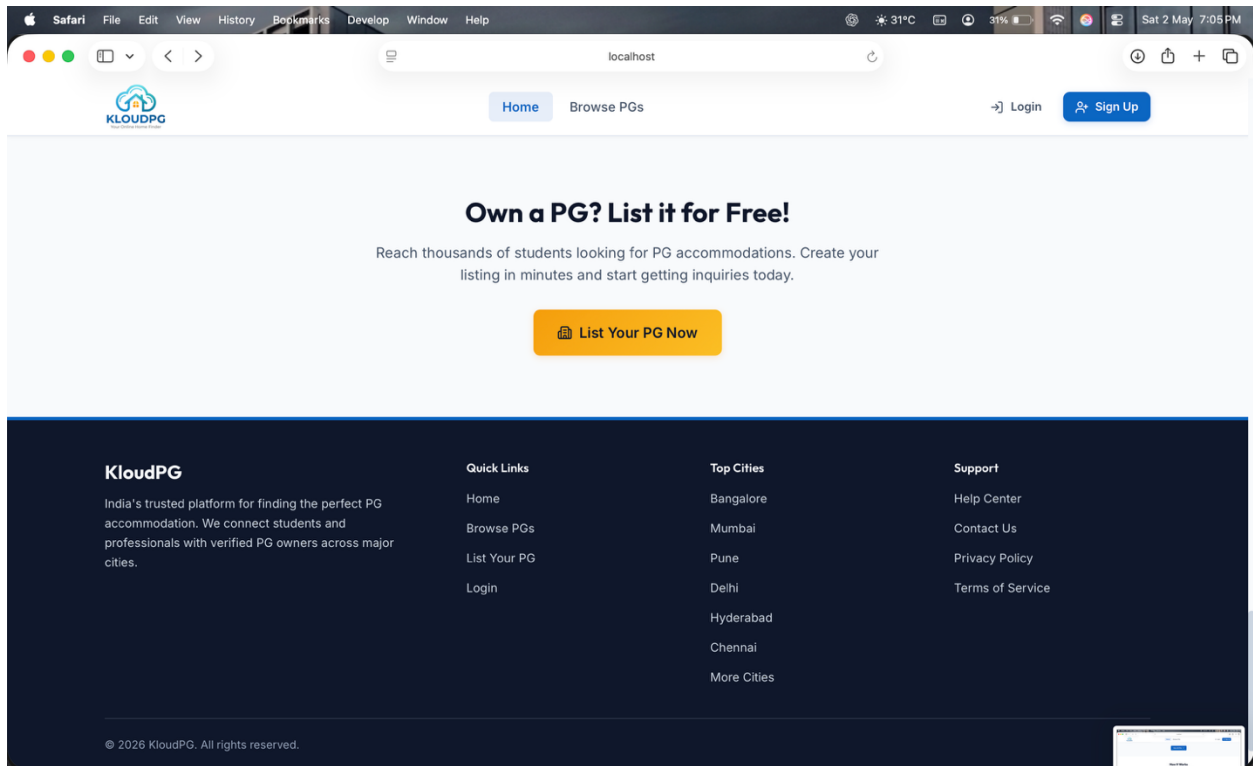
By executing these test cases, we verify that the **Business Logic** (Class Diagram), **Functional Flow** (Activity Diagram), and **Data Integrity** (Database Design) work together seamlessly to provide a secure and efficient user experience.

8. Screenshots of Live Project Demo

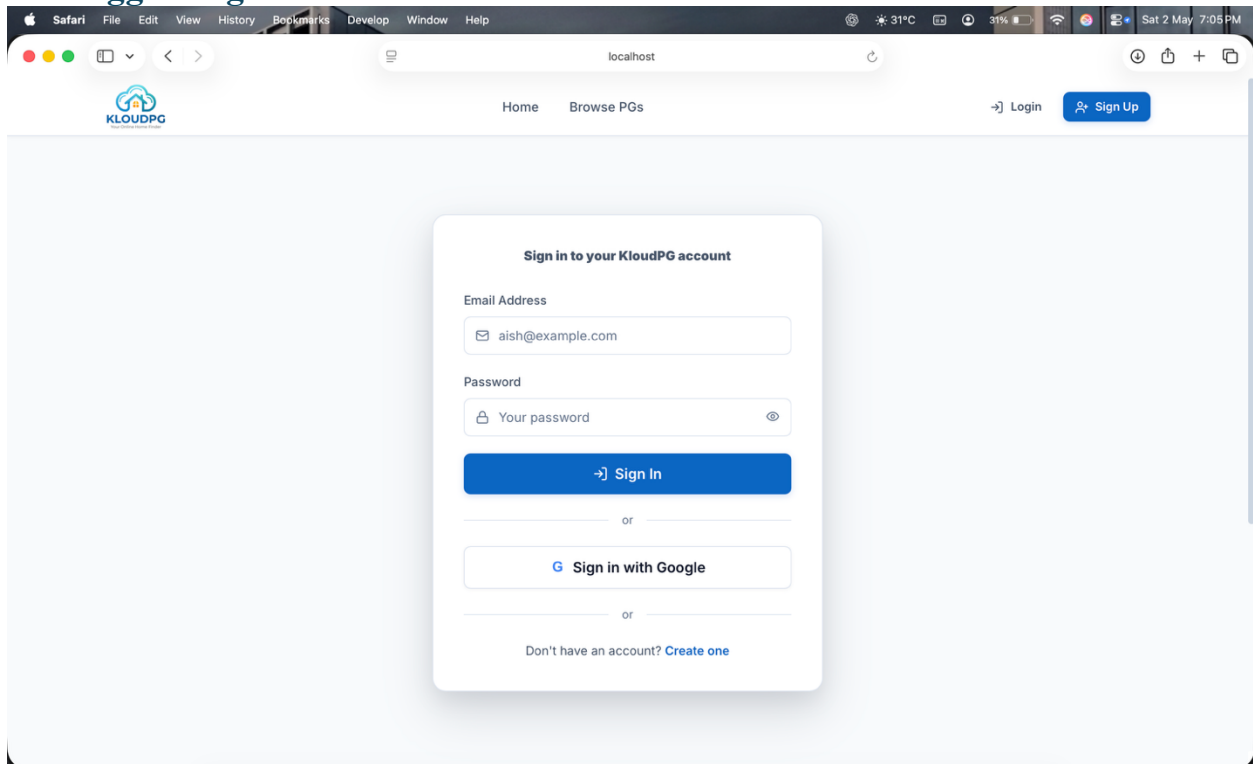
8.1 Main Page

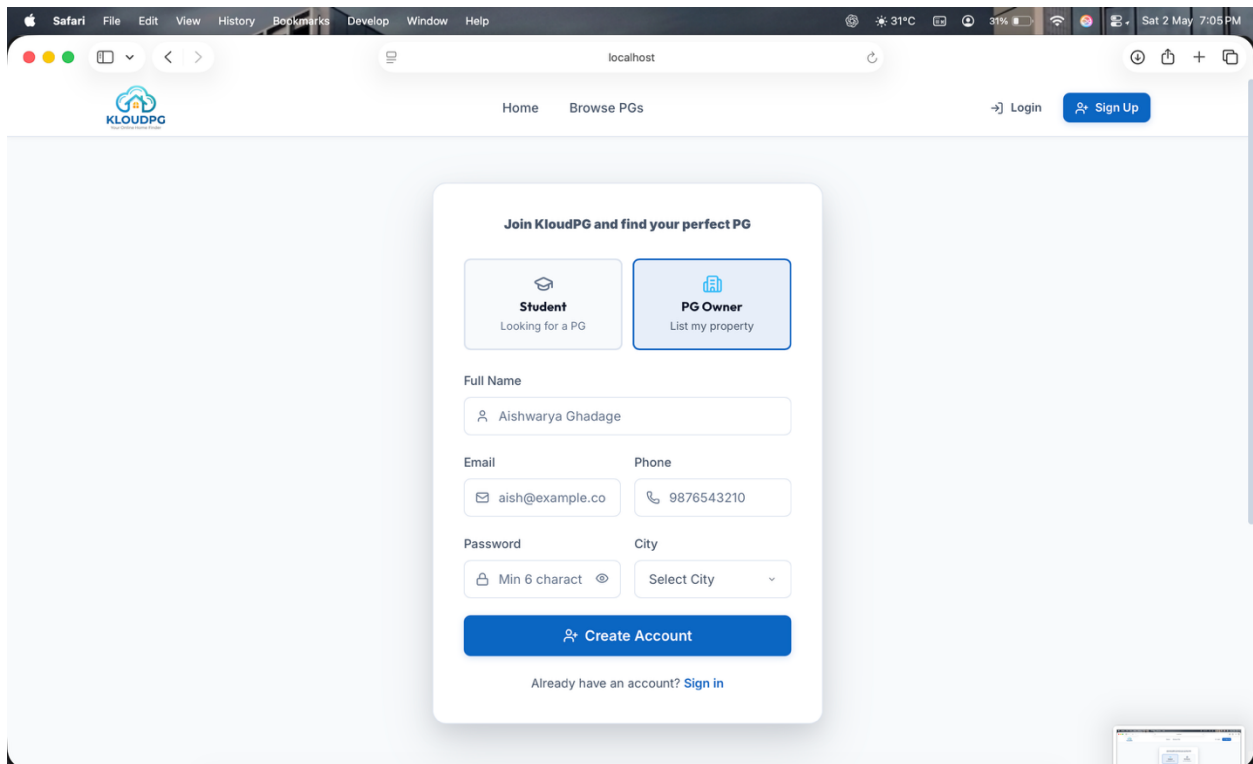
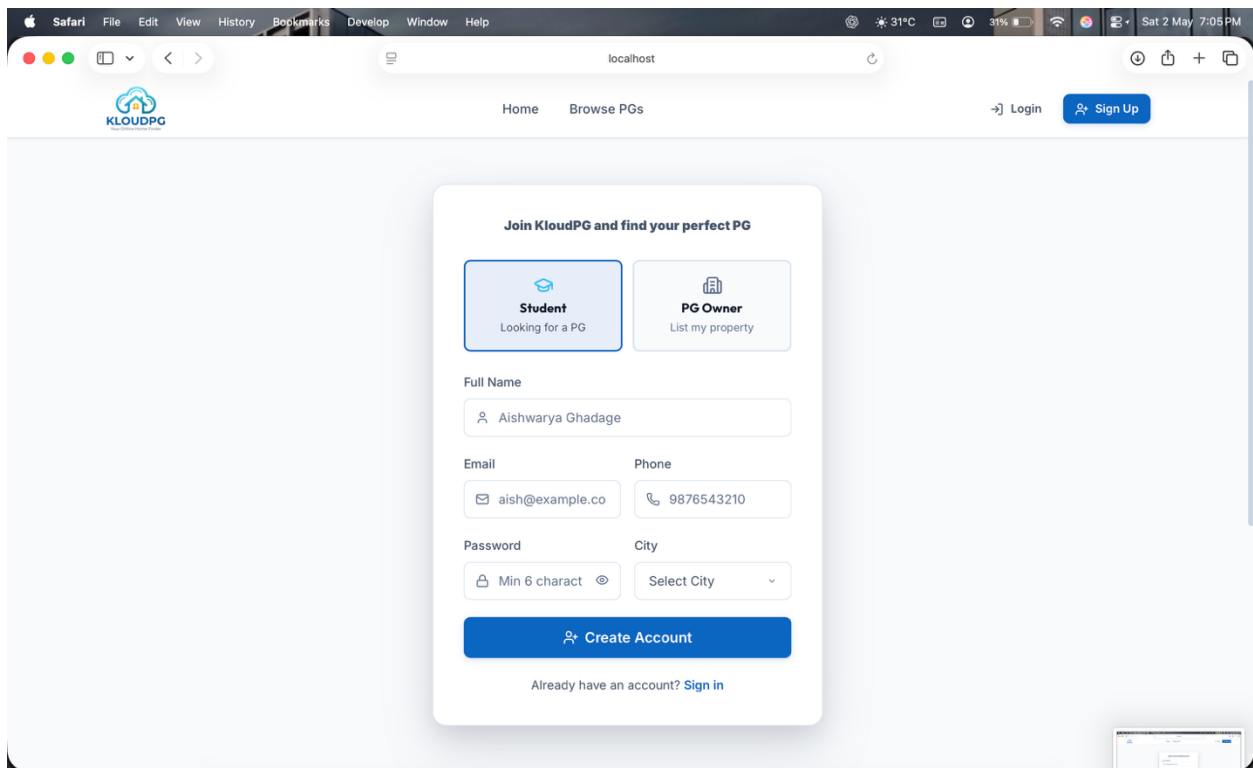




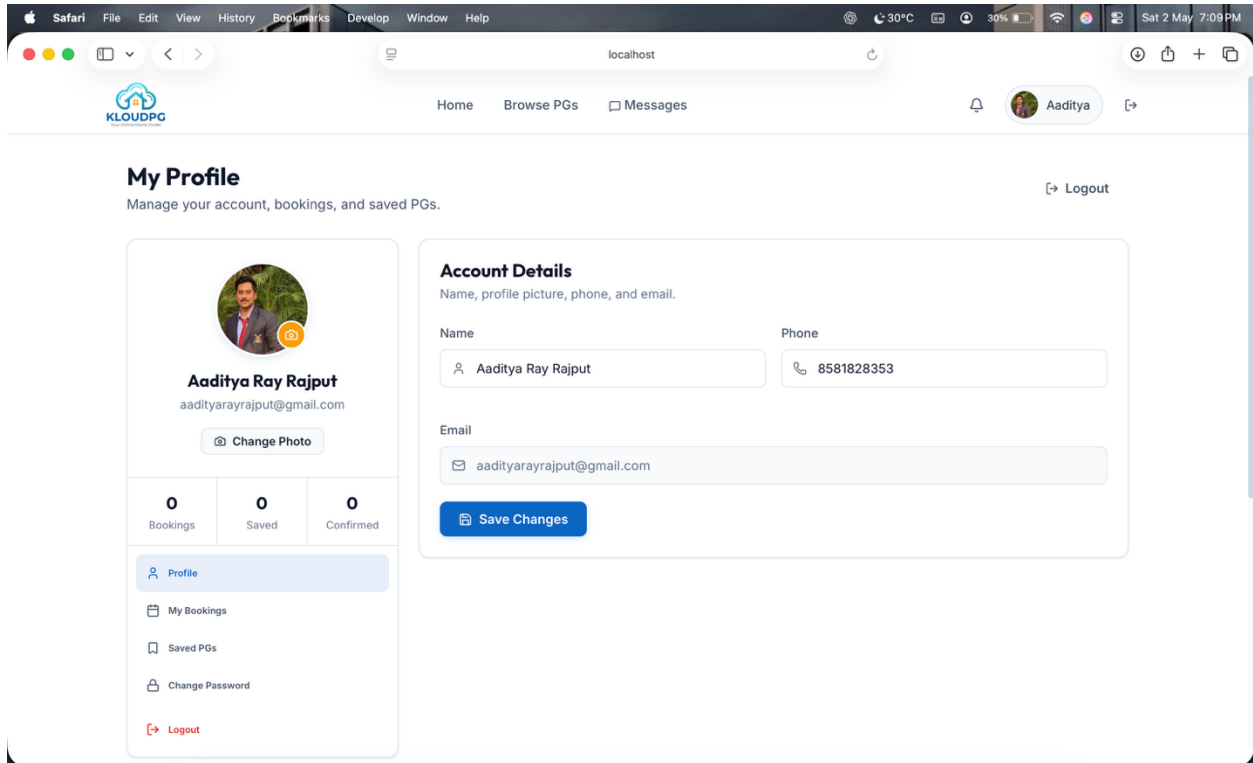


8.2 Loggin Page



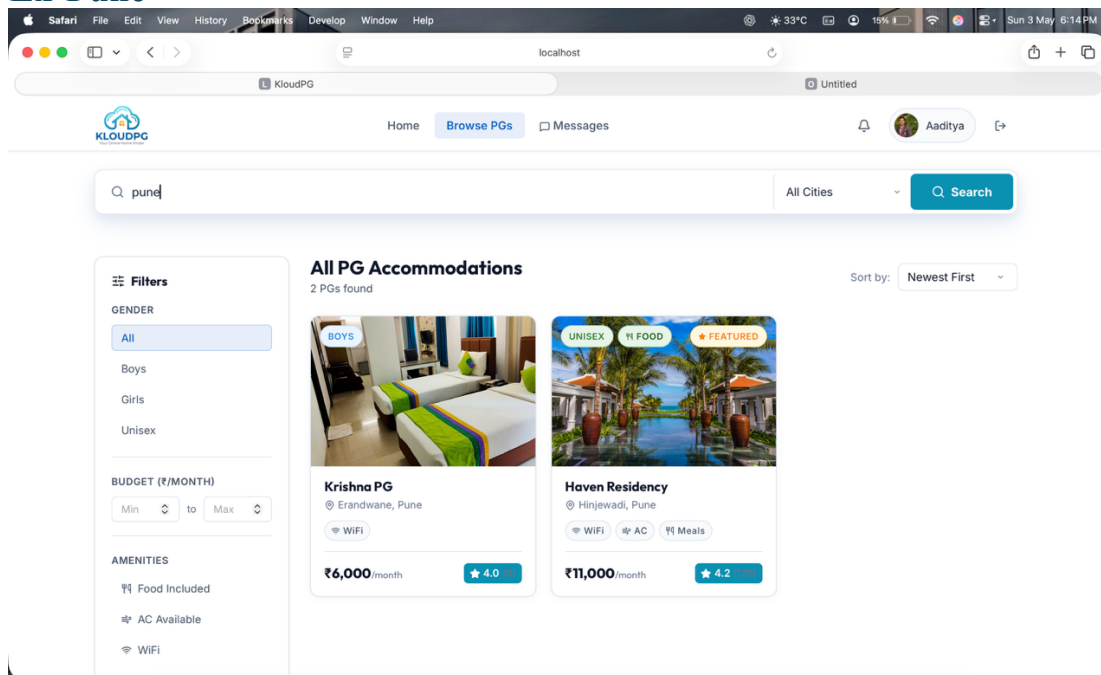


8.3 Profile Dashboard



8.4 Search Bar

8.4.1 Search Pg Ex-Pune



8.4.2 Filter



HomeBrowse PGsMessages

Aishwarya

13 PGs found

Filters

GENDER

AllBoysGirlsUnisex

BUDGET (₹/MONTH)

Min to Max

AMENITIES

☒ Food Included

☒ AC Available

☒ WiFi

Clear All Filters

BOYS **FOOD**



Krishna PG
Erandwane, Pune
WiFi Meals
₹6,000/month **4.0**

BOYS **FOOD**



Silver Quarters
T. Nagar, Chennai
WiFi AC Meals
₹10,000/month **4.4**

BOYS



Happy House
Adyar, Chennai
WiFi
₹6,000/month **3.5**

UNISEX **FOOD**



Urban Lodge
Kukatpally, Hyderabad

GIRLS **FOOD**



Star Homes
Dwarka, Delhi

UNISEX **FOOD** **FEATURED**



Haven Residency
Hinjewadi, Pune



HomeBrowse PGsMessages

Aishwarya

All PG Accommodations

13 PGs found

Sort by: Newest First

Filters

GENDER

AllBoysGirlsUnisex

BUDGET (₹/MONTH)

Min to Max

AMENITIES

☒ Food Included

☒ AC Available

☒ WiFi

Clear All Filters

BOYS **FOOD**



Krishna PG
Erandwane, Pune
WiFi Meals
₹6,000/month **4.0**

BOYS **FOOD**



Silver Quarters
T. Nagar, Chennai
WiFi AC Meals
₹10,000/month **4.4**

BOYS



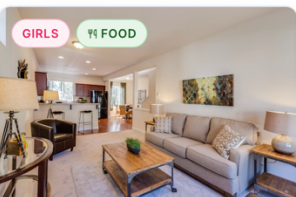
Happy House
Adyar, Chennai
WiFi
₹6,000/month **3.5**

UNISEX **FOOD**



Urban Lodge

GIRLS **FOOD**



Star Homes

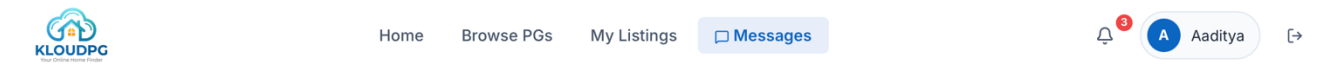
UNISEX **FOOD** **FEATURED**



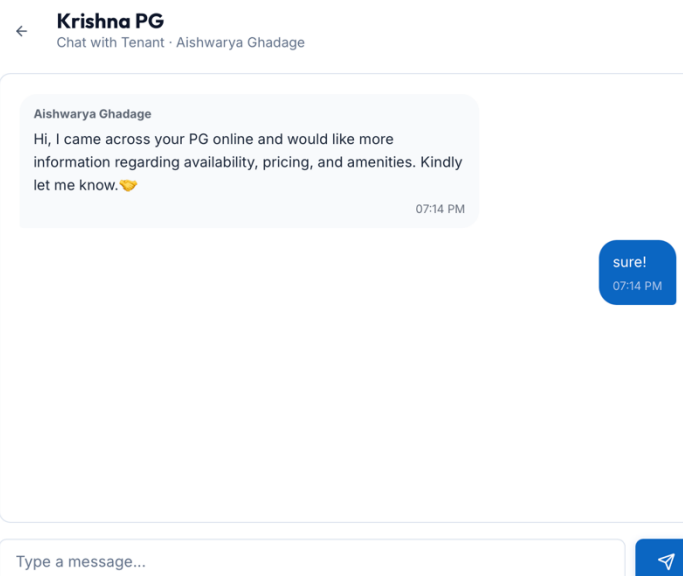
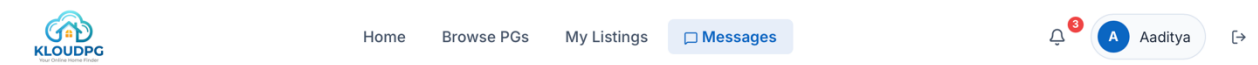
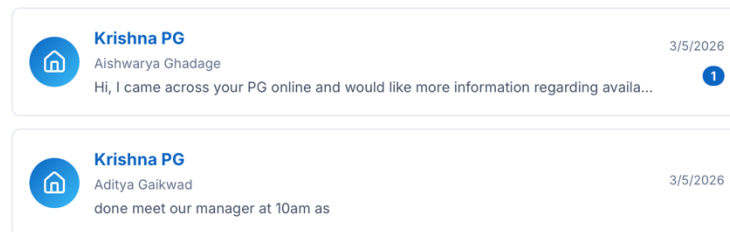
Haven Residency

8.5 Message Section

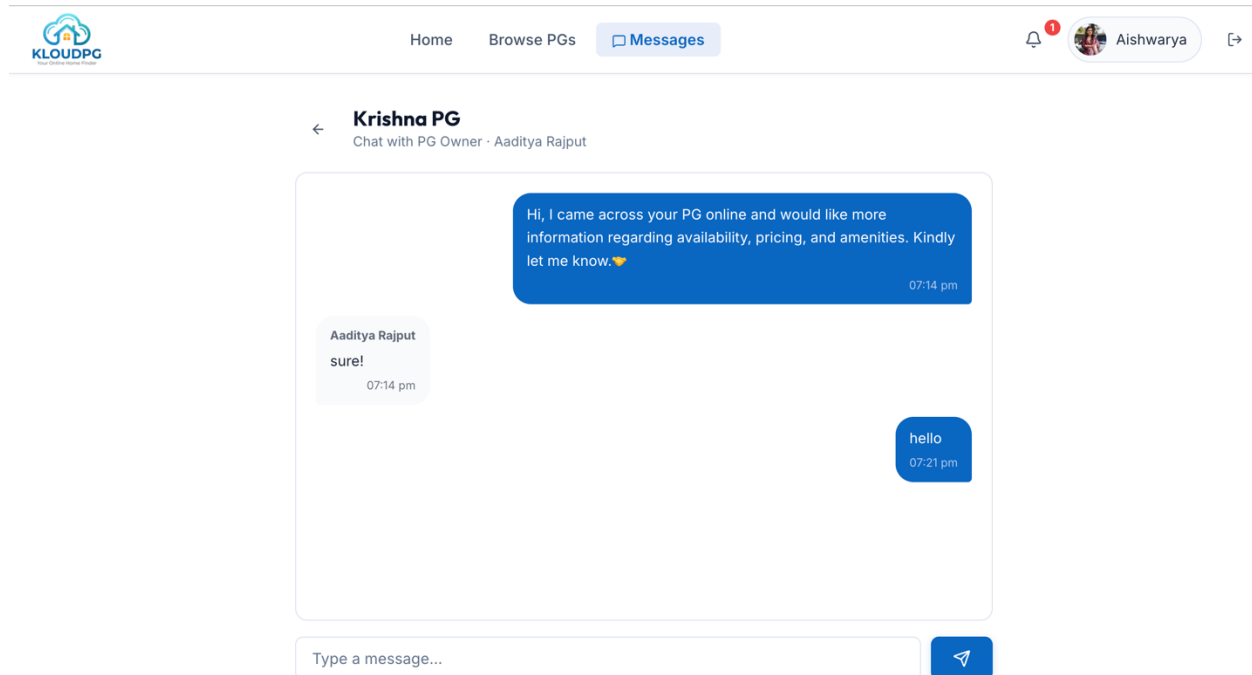
8.5.1 Owner End



Messages

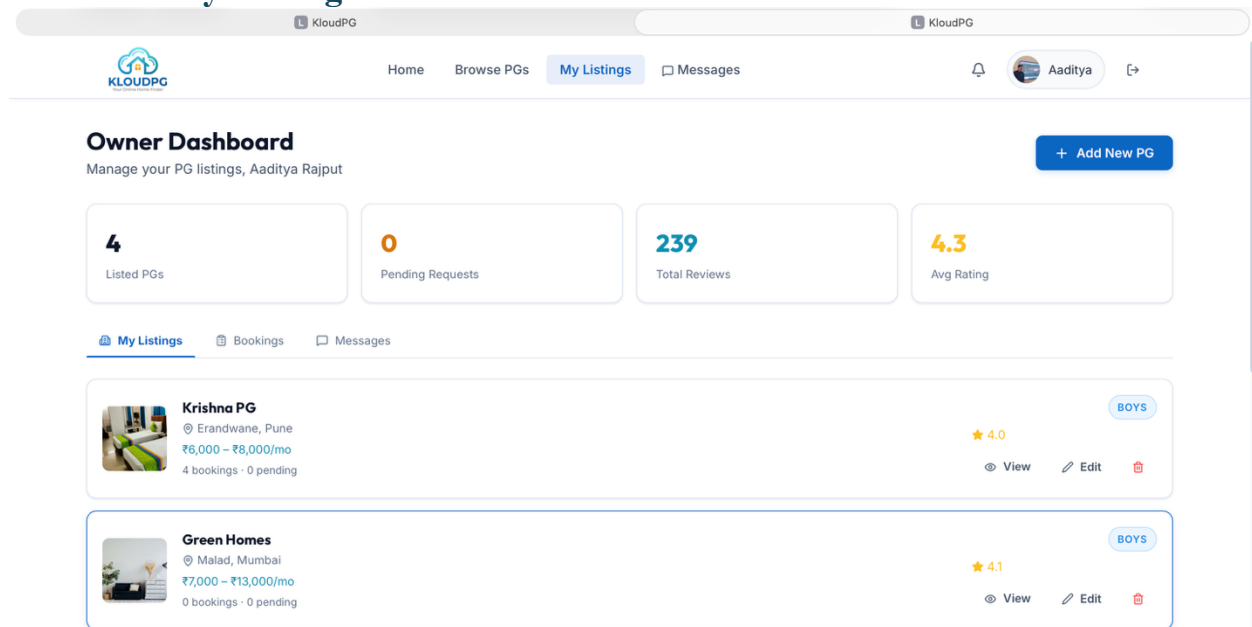


8.5.2 User End



8.6 Owner Dashboard

8.6.1 My Listing



8.6.2 Bookings

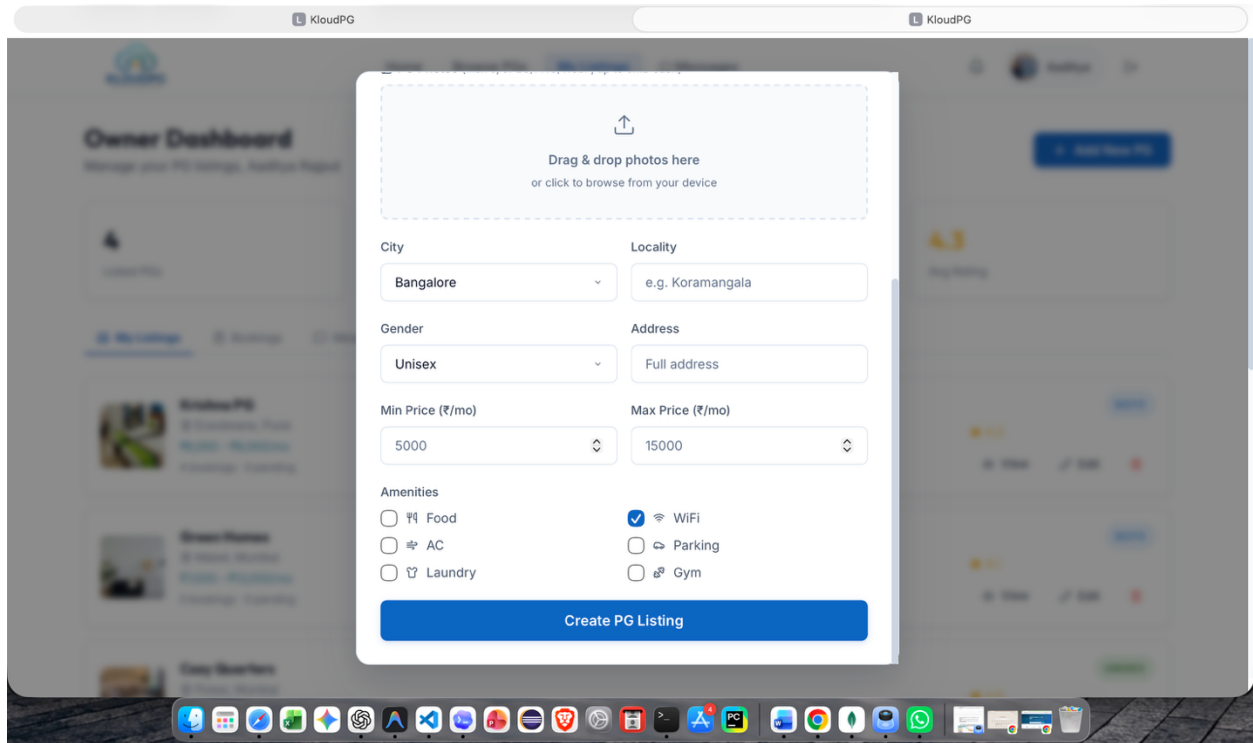
The screenshot shows the 'Bookings' tab in the KloudPG dashboard. The header includes navigation links: Home, Browse PGs, My Listings (active), and Messages. The user profile 'Aaditya' is visible in the top right. The bookings list contains the following entries:

Guest Name	Status	Contact Info	Room Type	Check-in	Check-out	Action
Aditya Gaikwad	CONFIRMED	adityag17@gmail.com · 7040040527	Double Room		3/5/2026	Chat
Aishwarya Ghadage	CANCELLED	aishwaryag15@gmail.com · 8010222627	Double Room	2026-05-16	2/5/2026	Chat
Aishwarya Ghadage	CANCELLED	aishwaryag15@gmail.com · 8010222627	Double Room	2026-05-20	2/5/2026	Chat
Rohan Kale	CONFIRMED	rohankale0267@gmail.com · 9284825624	Single Room	2026-05-04	1/5/2026	Chat

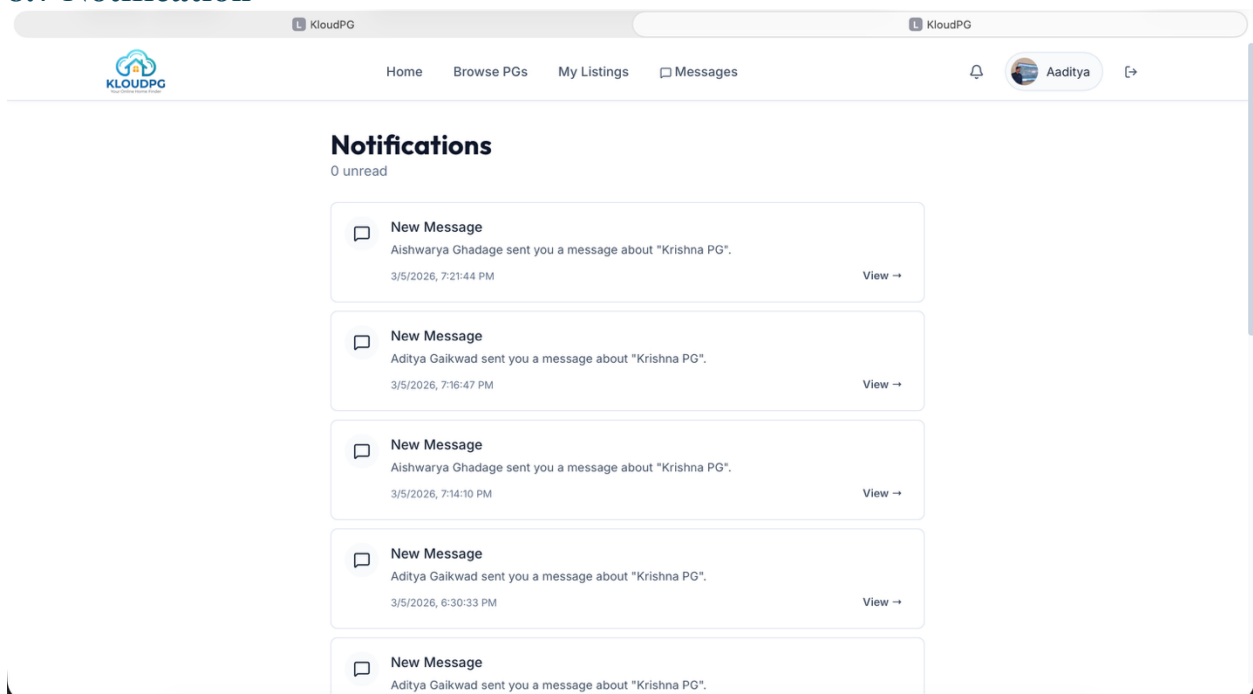
8.6.3 Add New Pg

The screenshot shows the 'Add New PG' modal form. The form includes the following fields and options:

- PG Name:** Text input field with placeholder 'e.g. Sunshine PG'.
- Description:** Text area with placeholder 'Describe your PG...'.
- PG Photos:** Checkmark icon, text '(max 5, JPEG/PNG/WebP, up to 5MB each)', and a dashed box with an upload icon and text 'Drag & drop photos here or click to browse from your device'.
- City:** Dropdown menu with 'Bangalore' selected.
- Locality:** Text input field with placeholder 'e.g. Koramangala'.
- Gender:** Text input field.
- Address:** Text input field.



8.7 Notification



8.8 Admin Panel

8.8.1 Admin Dashboard

HomeBrowse PGs**Admin**MessagesAdmin

Admin Dashboard

System overview, Admin

UsersPGsBookings**Complaints**Reports

**17**
Total Users

**13**
Total PGs

**0**
Pending Approval

**6**
Total Bookings

**1**
Pending Bookings

**₹0**
Total Revenue

**0**
Open Complaints

**0**
Avg Rating

Quick Actions

**Approve PG Listings**
0 pending

**Manage Users**
17 total users

**Resolve Complaints**
0 open

**Monitor Bookings**
1 pending

**Generate Reports**
Export performance CSV

Bookings (Last 7 Days)

8.8.2 Pg listing control

HomeBrowse PGs**Admin**MessagesAdmin

PG Listings Control

Approve or reject PG listings submitted by owners

AllPendingApproved

PG NAME

OWNER

CITY

PRICE

BOOKINGS

RATING

STATUS

ACTIONS

Krishna PG Erandwane	Aaditya Rajput owner1@demo.com	Pune	₹6,000-8,000	4	★ 4.0	Approved	 Chat Owner  Reject
Silver Quarters T. Nagar	Suresh Iyer owner9@demo.com	Chennai	₹10,000-14,000	0	★ 4.4	Approved	 Chat Owner  Reject
Happy House Adyar	Vikram Singh owner5@demo.com	Chennai	₹6,000-11,000	0	★ 3.5	Approved	 Chat Owner  Reject
Urban Lodge Kukatpally	Ramesh Rao owner11@demo.com	Hyderabad	₹11,000-13,000	0	★ 4.2	Approved	 Chat Owner  Reject
Star Homes Dwarka	Ramesh Rao owner11@demo.com	Delhi	₹9,000-13,000	0	★ 4.5	Approved	 Chat Owner  Reject
Haven Residency Hinjewadi	Deepa Gupta owner10@demo.com	Pune	₹11,000-13,000	0	★ 4.2	Approved	 Chat Owner  Reject
Green Homes Malad	Aaditya Rajput owner1@demo.com	Mumbai	₹7,000-13,000	0	★ 4.1	Approved	 Chat Owner  Reject

8.8.3 User Management



HomeBrowse PGsAdminMessages

Admin

User Management

View and manage all registered users

Admin Dashboard

18

Total Users

5

Students

12

Owners

0

Blocked

Search by name or email...

All Roles

Search

NAME	EMAIL	ROLE	CITY	BOOKINGS	REVIEWS	JOINED	STATUS	ACTION
Aaditya Ray Rajput	aadityarayrajput@gmail.com	Student	Patna	0	0	2/5/2026	Active	Block
Aishwarya Ghadage	aishwaryag15@gmail.com	Student	Pune	2	0	1/5/2026	Active	Block
Aditya Gaikwad	adityag17@gmail.com	Student	Pune	1	0	1/5/2026	Active	Block
Rohan Kale	rohankale0267@gmail.com	Student	Pune	1	0	1/5/2026	Active	Block
Lakshmi Pillai	owner12@demo.com	Owner	Bangalore	0	0	1/5/2026	Active	Block
Ramesh Rao	owner11@demo.com	Owner	Bangalore	0	0	1/5/2026	Active	Block

8.8.4 Overall Report of Webapp (KloudPg)



HomeBrowse PGsAdminMessages

Admin

Reports

Track platform performance and export admin reports

BackExport CSV

7 Days30 Days90 Days

₹0

Revenue in range

0

Completed payments

6

Total bookings

0

Average rating

Booking Status

Confirmed3

Cancelled2

Pending1

Complaint Status

PGs by City

CITY	TOTAL PGS	APPROVE

Top PGs

Krishna PG

Pune

4

8.9 Database (Stored Data in JSON Format)

8.9.1 Pgs

KloudPg > kloudpg > pgs Open MongoDB shell

Documents 13 Aggregations Schema Indexes 4 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE 25 1 - 13 of 13 ⏪ ⏩ ⌵ ☰ {} ⌂

```
_id: ObjectId('69f44bafaad45d8ee849160e')
id: 1
owner_id: 11
name: "Golden Stay"
description: "Premium PG accommodation with AC rooms, hot water, and power backup. 0..."
city: "Bangalore"
locality: "Koramangala"
address: "294, Koramangala, Bangalore"
gender: "boys"
food_included: 0
wifi: 1
ac: 1
parking: 1
laundry: 0
gym: 0
power_backup: 1
cctv: 1
water_purifier: 1
price_min: 9000
price_max: 15000
rating: 3.9
total_reviews: 47
is_featured: 1
is_approved: 1
image_url: "https://images.unsplash.com/photo-1522771739844-6a9f6d5f14af?w=600&q=8..."
```

[Show 1 more field](#)

8.9.2 pg-rooms

KloudPg > kloudpg > pg_rooms Open MongoDB shell

Documents 39 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE 25 1 - 25 of 39 ⏪ ⏩ ⌵ ☰ {} ⌂

```
_id: ObjectId('69f44bafaad45d8ee849160f')
id: 1
pg_id: 1
room_type: "single"
price: 15000
available_count: 2
created_at: 2026-05-01T06:43:59.436+00:00
```

```
_id: ObjectId('69f44bafaad45d8ee8491610')
id: 2
pg_id: 1
room_type: "double"
price: 12000
available_count: 5
created_at: 2026-05-01T06:43:59.462+00:00
```

```
_id: ObjectId('69f44bafaad45d8ee8491611')
id: 3
pg_id: 1
room_type: "triple"
price: 9000
available_count: 4
created_at: 2026-05-01T06:43:59.462+00:00
```

```
_id: ObjectId('69f44bafaad45d8ee8491627')
id: 10
pg_id: 4
room_type: "single"
price: 14000
available_count: 6
```

8.9.3 Booking

KloudPg > kloudpg > bookings

>_ Open MongoDB shell

Documents 6 Aggregations Schema Indexes 4 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25

1 - 6 of 6

↺

↻

↷

⋮

⌘

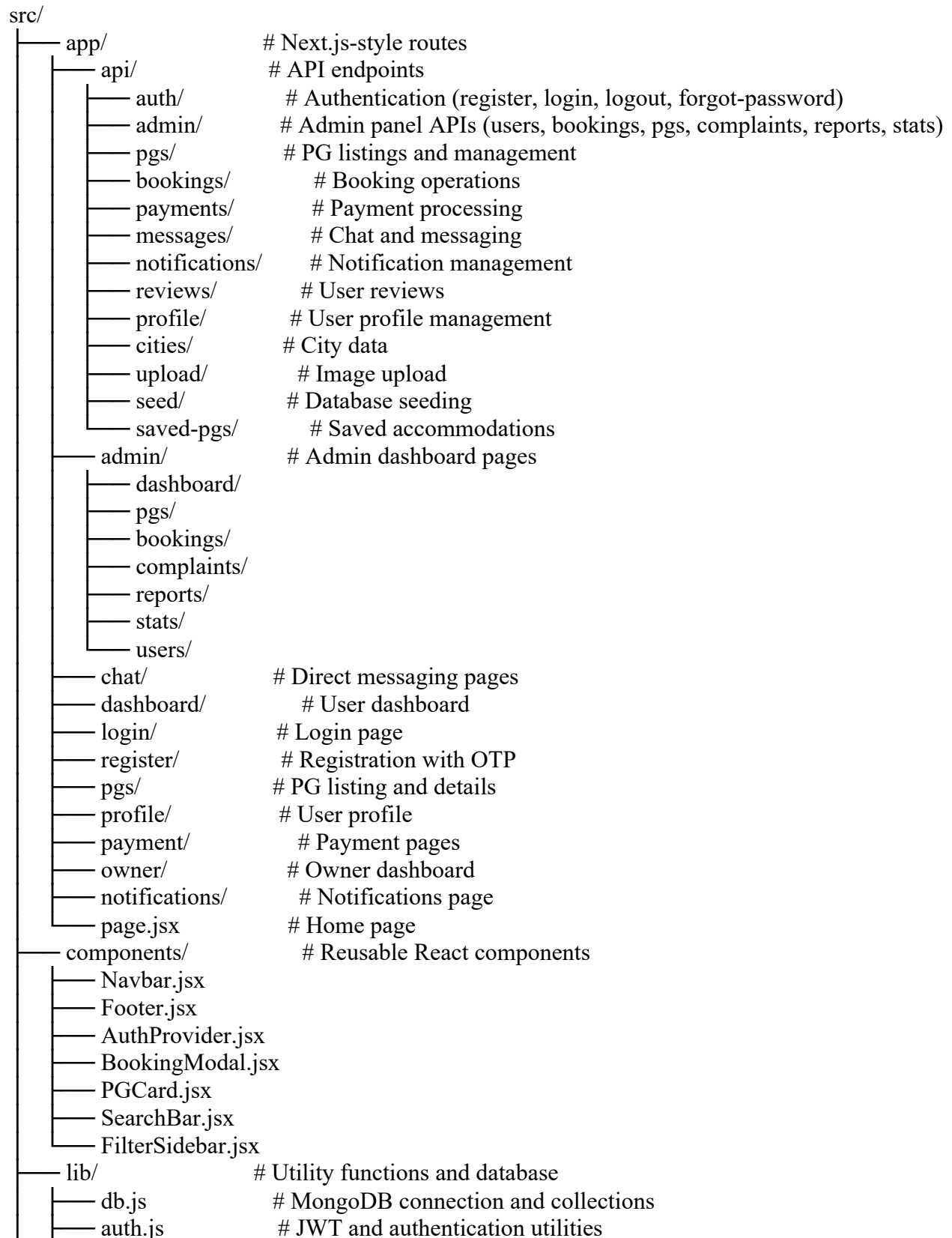
|

⌘

```
_id: ObjectId('69f44bafaad45d8ee84916e6')
id: 1
user_id: 1
pg_id: 1
room_type: "double"
status: "confirmed"
visit_date: "2026-04-10"
message: "I am a student at IIT Bangalore, looking for PG near campus."
phone: "9876543210"
created_at: 2026-05-01T06:43:59.560+00:00

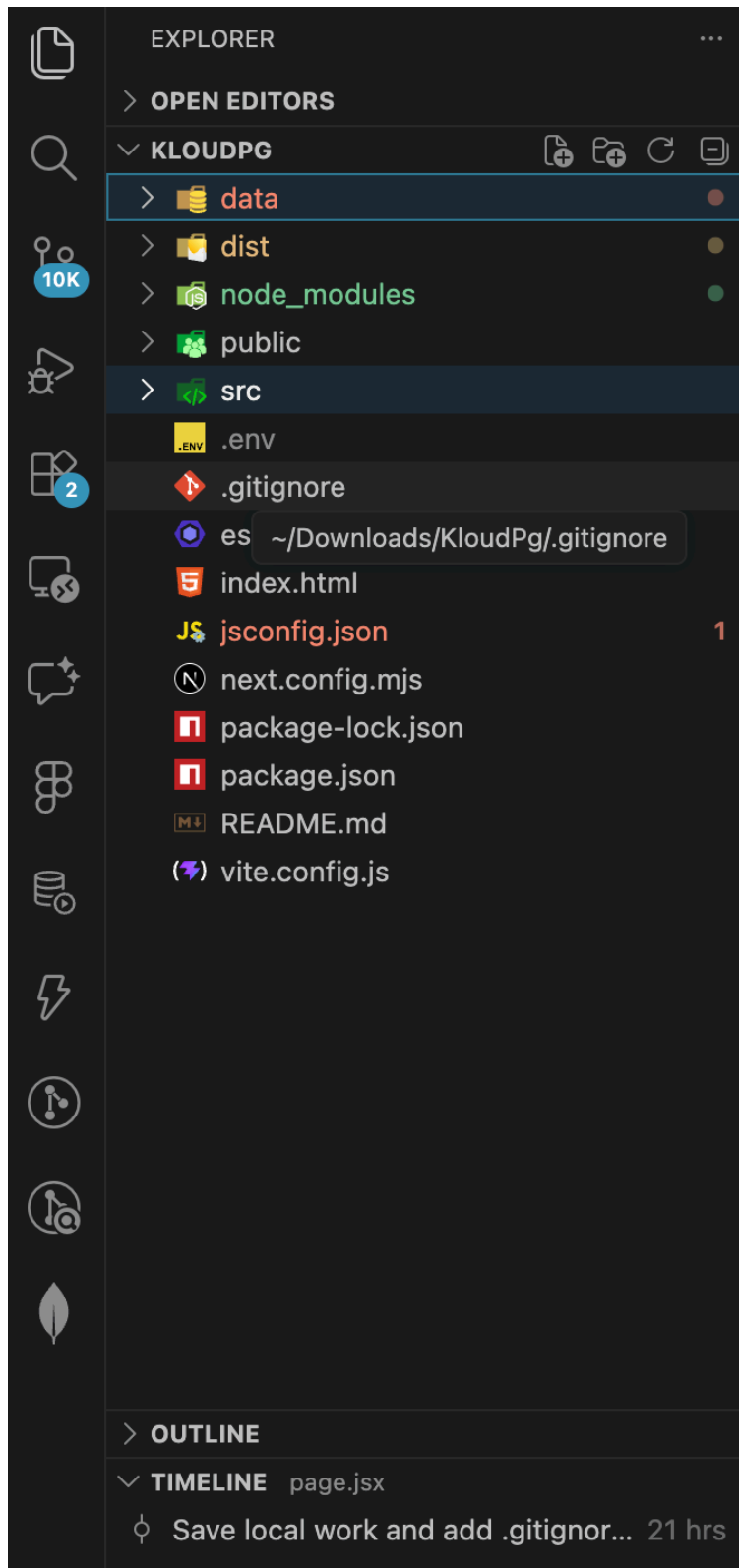
_id: ObjectId('69f44bafaad45d8ee84916e7')
id: 2
user_id: 1
pg_id: 3
room_type: "single"
status: "pending"
visit_date: "2026-04-15"
message: "Looking for a single room with AC. Will be joining office next month."
phone: "9876543210"
created_at: 2026-05-01T06:43:59.560+00:00
```

8.9.4 Project Structure

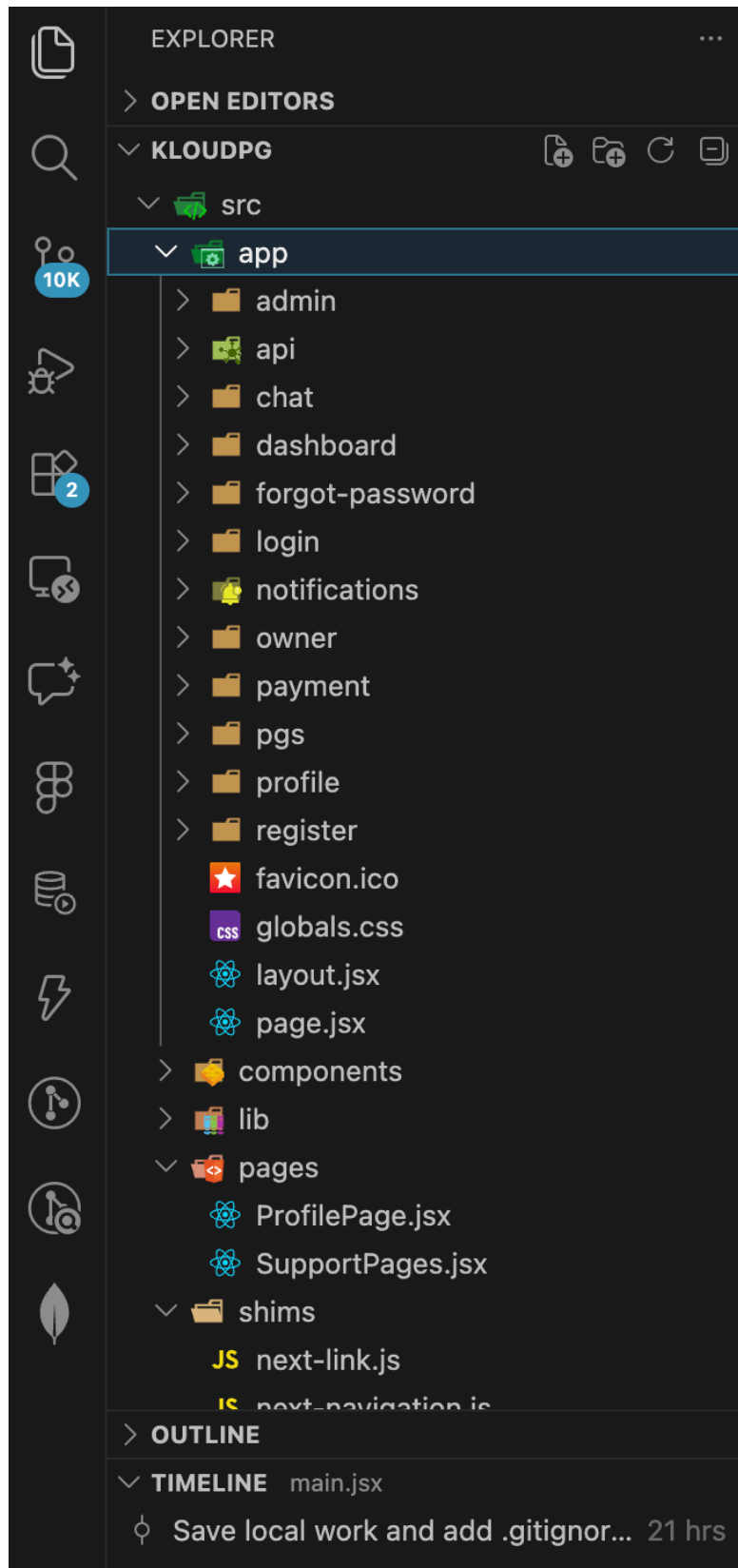


```
|
|├── email.js          # Email sending with Nodemailer
|├── cities.js         # City data
|├── seed.js           # Database seeding
|├── migrate-images.mjs # Image migration utility
|├── pages/            # Additional page components
|│   ├── ProfilePage.jsx
|│   └── SupportPages.jsx
|├── shims/            # Next.js compatibility shims for Vite
|│   ├── next-link.js
|│   ├── next-navigation.js
|│   └── next-server.js
|├── App.jsx           # Main React component with routing
└── main.jsx           # Application entry point
```

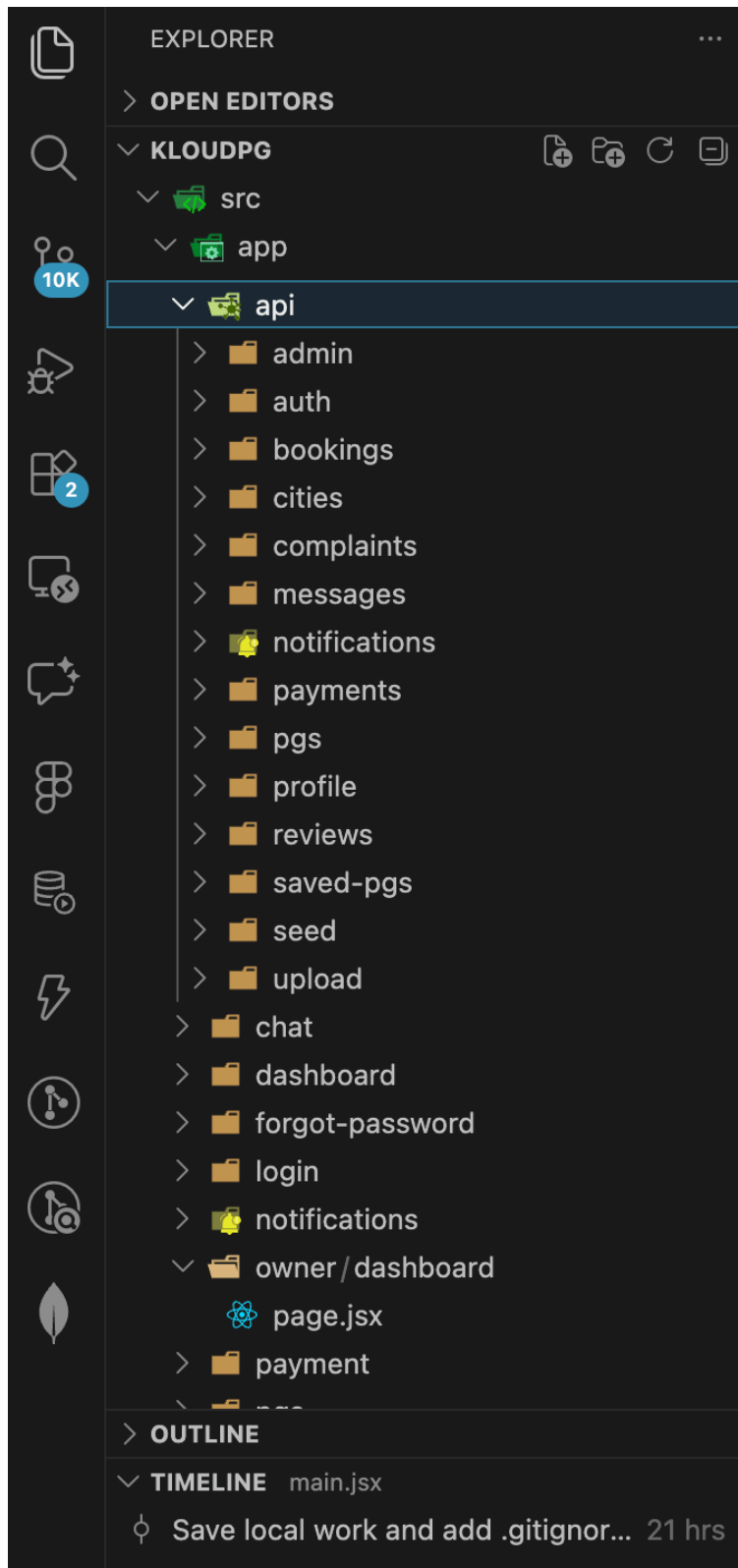
Directory	Purpose
 api	Backend API endpoints (40+ routes)
 app (non-api)	Frontend pages & layouts
 components	Reusable React components (7 files)
 lib	Shared utilities (db, auth, seed)
 pages	Support pages
 data	Local SQLite/MongoDB database
 uploads	User-uploaded PG images



Overview Of Bankend and Frontend



Backend



Frontend

8.10 Appendix Features

8.10.1 Help Centre

HomeBrowse PGsMessages

⁶Aishwarya→



Help Center

Find quick answers for bookings, saved PGs, owner listings, and payments.

How do I book a PG visit?

Open any PG listing, choose Book Visit, fill in your visit date and phone number, then submit the request.

Where can I see my bookings?

After login, open Profile or Dashboard and select My Bookings to track pending and confirmed requests.

How do I save a PG?

Use the Save action on a PG listing. Saved PGs appear in your Profile under Saved PGs.

Can owners list PGs?

Yes. Register as an owner, open My Listings, and add your PG details with photos, pricing, and amenities.

Ready to continue?[Browse PGs](#)

8.10.2 Contact Us

HomeBrowse PGsMessages

⁶Aishwarya→



Contact Us

Need help with a booking or listing? Reach the KloudPG support team.

**Email**
support@kloudpg.com

**Phone**
+91 9006352900

**Office**
Viman Nagar, Pune, India

Ready to continue?[Browse PGs](#)

40

8.10.2 Similar suggestion



HomeBrowse PGsMessages



Aishwarya

Similar PGs in Pune

Other accommodations you might like

UNISEXFOODFEATURED



Haven Residency

Hinjewadi, Pune

WiFiACMeals

₹11,000/month4.2 (125)

Filter Service:



Filters

GENDER

AllBoysGirlsUnisex

BUDGET (₹/MONTH)

Min to Max

AMENITIES

Food Included

AC Available

WiFi

Clear All Filters

Footer:

KloudPG India's trusted platform for finding the perfect PG accommodation. We connect students and professionals with verified PG owners across major cities.	Quick Links Home Browse PGs List Your PG Login	Top Cities Bangalore Mumbai Pune Delhi Hyderabad Chennai More Cities	Support Help Center Contact Us Privacy Policy Terms of Service
---	---	--	---

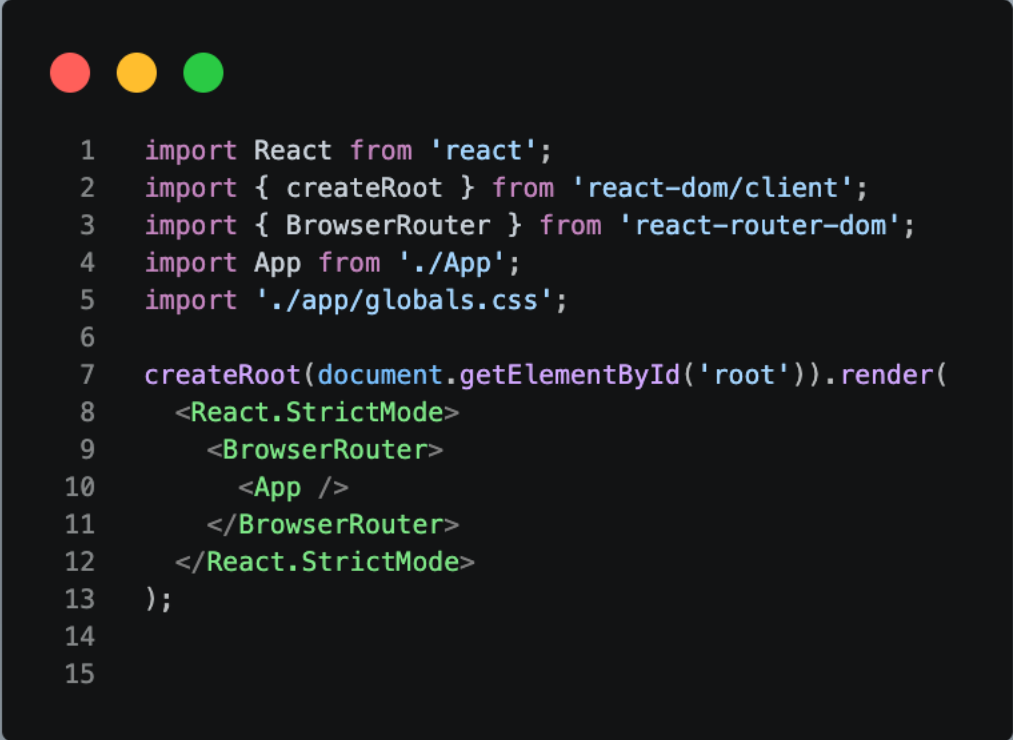
© 2026 KloudPG. All rights reserved.

8.11 Main module codes

8.11.1 App.jsx

```
1  import { Routes, Route } from 'react-router-dom';
2  import { AuthProvider } from '@components/AuthProvider';
3  import Navbar from '@components/Navbar';
4  import Footer from '@components/Footer';
5
6  import HomePage from '@app/page';
7  import LoginPage from '@app/login/page';
8  import RegisterPage from '@app/register/page';
9  import ForgotPasswordPage from '@app/forgot-password/page';
10 import PGsPage from '@app/pgs/page';
11 import PGDetailPage from '@app/pgs/[id]/page';
12 import DashboardPage from '@app/dashboard/page';
13 import OwnerDashboardPage from '@app/owner/dashboard/page';
14 import AdminDashboardPage from '@app/admin/dashboard/page';
15 import AdminUsersPage from '@app/admin/users/page';
16 import AdminPGsPage from '@app/admin/pgs/page';
17 import AdminBookingsPage from '@app/admin/bookings/page';
18 import AdminComplaintsPage from '@app/admin/complaints/page';
19 import AdminReportsPage from '@app/admin/reports/page';
20 import NotificationsPage from '@app/notifications/page';
21 import ChatInboxPage from '@app/chat/page';
22 import ChatPage from '@app/chat/[pg_id]/page';
23 import PaymentPage from '@app/payment/[booking_id]/page';
24 import ProfilePage from '@pages/ProfilePage';
25 import { ContactPage, HelpPage, PrivacyPage, TermsPage } from '@pages/SupportPages';
26
27 export default function App() {
28   return (
29     <AuthProvider>
30       <Navbar />
31       <main className="page-wrapper">
32         <Routes>
33           <Route path="/" element={<HomePage />} />
34           <Route path="/login" element={<LoginPage />} />
35           <Route path="/register" element={<RegisterPage />} />
36           <Route path="/forgot-password" element={<ForgotPasswordPage />} />
37           <Route path="/pgs" element={<PGsPage />} />
38           <Route path="/pgs/:id" element={<PGDetailPage />} />
39           <Route path="/dashboard" element={<DashboardPage />} />
40           <Route path="/owner/dashboard" element={<OwnerDashboardPage />} />
41           <Route path="/admin/dashboard" element={<AdminDashboardPage />} />
42           <Route path="/admin/users" element={<AdminUsersPage />} />
43           <Route path="/admin/pgs" element={<AdminPGsPage />} />
44           <Route path="/admin/bookings" element={<AdminBookingsPage />} />
45           <Route path="/admin/complaints" element={<AdminComplaintsPage />} />
46           <Route path="/admin/reports" element={<AdminReportsPage />} />
47           <Route path="/notifications" element={<NotificationsPage />} />
48           <Route path="/chat" element={<ChatInboxPage />} />
49           <Route path="/chat/:pg_id" element={<ChatPage />} />
50           <Route path="/payment/:booking_id" element={<PaymentPage />} />
51           <Route path="/profile" element={<ProfilePage />} />
52           <Route path="/help" element={<HelpPage />} />
53           <Route path="/contact" element={<ContactPage />} />
54           <Route path="/privacy" element={<PrivacyPage />} />
55           <Route path="/terms" element={<TermsPage />} />
56         </Routes>
57       </main>
58       <Footer />
59     </AuthProvider>
60   );
61 }
62
```

8.11.2 App.jsx



```
1  import React from 'react';
2  import { createRoot } from 'react-dom/client';
3  import { BrowserRouter } from 'react-router-dom';
4  import App from './App';
5  import './app/globals.css';
6
7  createRoot(document.getElementById('root')).render(
8    <React.StrictMode>
9      <BrowserRouter>
10        <App />
11      </BrowserRouter>
12    </React.StrictMode>
13  );
14
15
```

9. USER MANUAL

This manual provides a step-by-step guide for the different users of the **Online PG Booking and Management System**, ensuring a smooth experience across all functional modules.

9.1 Customer Manual

The Customer role focuses on discovering, viewing, and securing accommodation.

- **Registration & Login:** New users must click "Sign Up" to create an account with their email and contact details. Existing users can log in using their credentials to access personalized features.
- **Searching for a PG:** Use the search bar on the homepage to enter a location (e.g., "Pune"). You can apply filters such as rent range, room type (Single/Sharing), and specific amenities like WiFi or AC.
- **Booking a Room:** Once a suitable PG is found, click "View Details" to check photos and availability. Select your check-in date and click "Book Now" to send a request to the owner.
- **Payment:** After the owner approves the request, navigate to "My Bookings" and click "Pay." Complete the transaction via the secure payment gateway using a card, UPI, or net banking.
- **Reviews & Feedback:** After your stay, you can navigate to the property page to leave a rating and review based on your experience.

9.2 PG Owner Manual

The Owner role is designed for managing property listings and coordinating with potential tenants.

- **Adding a property:** Click "Add New PG" on the dashboard. Fill in the property name, address, rent details, and upload high-quality images of the rooms.
- **Managing Room Availability:** Under the "Manage Rooms" section, owners can update the status of individual rooms to "Available" or "Booked" manually if needed.
- **Handling Booking Requests:** When a customer initiates a booking, a notification will appear in the "Requests" tab. Review the customer's profile and "Approve" or "Reject" the booking.

- **Tracking Earnings:** The "Earnings" tab provides a summary of all successful payments received and pending dues from active bookings.

9.3 Administrator Manual

The Admin role ensures the overall integrity, security, and smooth operation of the platform.

- **Approving Listings:** Admins must review newly added properties in the "Pending Approval" queue. Verify the details and images before making the PG live for customers.
- **User Management:** The Admin can view all registered Customers and Owners. They have the authority to block or de-activate accounts that violate system policies.
- **Conflict Resolution:** Access the "Messages" or "Complaints" section to mediate disputes between owners and customers regarding bookings or refunds.
- **System Reports:** Navigate to the "Reports" section to generate analytical data on total users, successful bookings, and monthly platform revenue.

9.4 Troubleshooting & Support

- **Payment Failure:** If a transaction fails, check your internet connection or bank balance. The system will retain the "Pending" status for a limited time to allow a retry.
- **Notification Issues:** Ensure that browser or app notifications are enabled in your account settings to receive real-time updates on bookings.
- **Password Recovery:** Use the "Forgot Password" link on the login page to receive a reset link via your registered email.

10. SCOPE AND LIMITATION

This section outlines the boundaries of the **Online PG Booking and Management System**, defining exactly what the platform achieves and the specific constraints it operates under.

10.1 SCOPE OF THE PROJECT

The scope defines the functional reach and the goals the system aims to fulfill:

- **Centralized Database:** To provide a unified platform where PG owners can list properties and customers can search for them without physical visits.
- **Real-Time Availability:** The system tracks room occupancy in real-time, preventing double bookings and ensuring customers see accurate information.
- **Secure Transactions:** Integration with payment gateways allows for safe and verifiable financial transactions between tenants and owners.
- **Automated Management:** Features like automated receipt generation, monthly revenue reports, and digital notifications reduce the manual workload for owners and admins.
- **Role-Based Access:** A clear hierarchy (Customer, Owner, Admin) ensures that each user has access only to the features relevant to them, maintaining security and privacy.

10.2 LIMITATIONS OF THE PROJECT

While the system is robust, certain constraints and boundaries exist:

- **Internet Dependency:** The platform is entirely web-based; users cannot search for or book PGs without an active internet connection.
- **Physical Verification:** While the system provides images and descriptions, it cannot guarantee the physical condition of the PG. Users are still encouraged to verify property details upon arrival.
- **Legal Formalities:** The system manages the booking and payment but does not currently handle the legal drafting or signing of rent agreements or police verification forms.
- **Hardware Constraints:** The performance of the system depends on the user's hardware (smartphone/PC) and the hosting server's capacity.

- **External API Reliance:** Functionality like maps (GPS) and payment processing depends on third-party services (like Google Maps or Razorpay). Any downtime from these providers will affect system features.

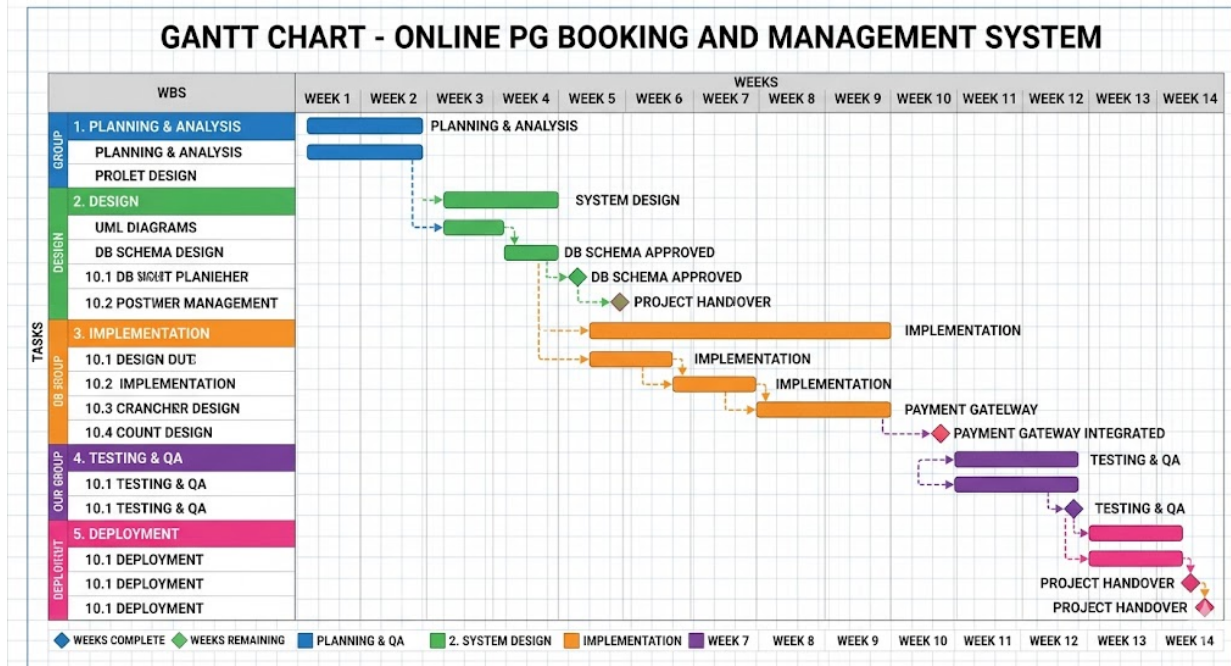
11. PROJECT SCHEDULING

Project scheduling is essential to ensure that the **Online PG Booking and Management System** is developed, tested, and deployed within the established timeframe.

11.1 GANTT CHART

A Gantt Chart provides a visual timeline of the project, showing the start and end dates of various phases. It helps in tracking progress and understanding the overlap between different tasks.

- **Requirement Analysis:** Gathering functional and non-functional requirements from stakeholders.
- **System Design:** Creating UML diagrams (Use Case, Class, Activity) and database schemas.
- **Implementation (Coding):** Developing the frontend (Web/Mobile) and backend (Node.js) logic.
- **Testing:** Executing module-wise and integration test cases to ensure system stability.
- **Deployment & Maintenance:** Hosting the application on a cloud server and providing ongoing support.

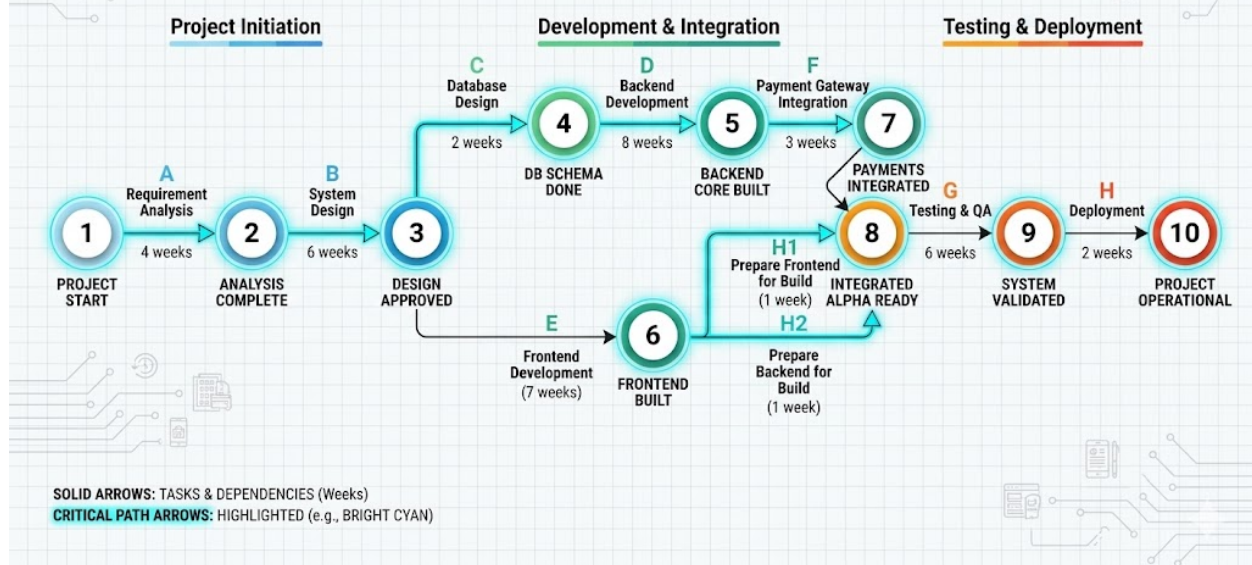


11.2 PERT CHART

The Program Evaluation and Review Technique (PERT) Chart focuses on the dependencies between tasks rather than just the timeline. It identifies the "Critical Path"—the sequence of stages determining the minimum time needed for project completion.

- **Task Dependencies:** The system design cannot begin until the requirement analysis is finalized.
- **Parallel Processing:** Backend development and database configuration can occur simultaneously once the design phase is complete.
- **Critical Milestones:** Successful integration of the Payment Gateway and Admin Approval modules are key milestones before moving to the final testing phase.

PERT CHART - ONLINE PG BOOKING AND MANAGEMENT SYSTEM



12. CONCLUSION

The development of the **Online PG Booking and Management System** addresses the critical need for a streamlined, transparent, and efficient way to manage paying guest accommodations. By integrating modern web technologies and a user-centric design, the project successfully bridges the gap between property owners and students or working professionals seeking reliable housing.

Key Achievements

- **Enhanced Accessibility:** The system provides a 24/7 digital platform, eliminating the need for physical searching and reducing the time required to secure accommodation.
- **Operational Efficiency:** Automated workflows for booking, payment processing, and notification management significantly reduce the manual administrative burden on PG owners.
- **Data Integrity and Security:** Through a robust database design and secure payment gateway integration, the platform ensures that user information and financial transactions are handled with high standards of safety.
- **Scalability:** The modular architecture—verified through comprehensive component and activity diagrams—allows for future enhancements, such as AI-based recommendations or multi-city expansion.

13. BIBLIOGRAPHY

The following resources, including textbooks, academic papers, and technical documentation, were consulted to ensure the technical accuracy and theoretical foundation of the **Online PG Booking and Management System**:

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